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EVENT AND OPPORTUNITY MANAGEMENT SYSTEM

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CHAPTER 1: SCOPE

1.1 System Overview

UniScout functions as an integrated web application created to streamline how campus participants identify and engage with various academic, professional, co-curricular, and personal growth endeavors. This solution aggregates campus-wide events curated by academic departments, faculties, student bodies, and registered organizations into a single unified directory.

participants may explore activities via interactive schedule views and searchable catalogues, filter choices by domain or eligibility parameters, bookmark selected activities, complete registrations, and receive automated alerts regarding upcoming dates and registration cutoffs.

Additionally, the environment provides tailored recommendations based on individual preferences, competencies, enrolled degree tracks, study levels, and prior engagement. Event hosts can publish and control activities, monitor participant sign-ups, issue announcements, and review attendance metrics. System administrators oversee entity verifications, review event submissions, manage accounts and taxonomies, and audit overall platform activities.

1.2 Problem Statement

Information regarding campus activities remains widely dispersed across departmental notices, club channels, social networks, and messaging threads. Consequently, participants frequently encounter difficulties discovering suitable events and tracking critical deadline dates.

Conventional linear or group-specific listings rarely assist participants in discovering options aligned with their specific interests, expertise, degree path, or academic standing. UniScout resolves this issue by consolidating activity listings and offering search tools, filtering, calendar navigation, sign-ups, saved items, reminders, and customized suggestions.

1.3 Objectives

The primary goal of UniScout is to offer a unified portal for discovering educational, professional, co-curricular, and skill-building activities. The portal allows participants to save options, submit registrations, and receive relevant alerts while delivering tailored suggestions derived from individual profiles and engagement history. Furthermore, UniScout provides organizers with tools to publish, administer, and track events and registrations, while equipping administrators with functionality to review listings, manage user access and classifications, and monitor site usage to minimize information fragmentation and promote active campus involvement.

1.4 Scope

1.4.1 Participant

- Perform account registration, login, logout, and credential recovery.
- Update participant ID, faculty affiliation, study program, academic year, personal interests, and skill sets.

- Explore, search, filter, and order available events.
- Access monthly, weekly, and daily calendar layouts.
- Sign up for activities or withdraw existing enrollments.
- Save interesting listings for future reference.
- Obtain alerts regarding schedules, cutoffs, event updates, and suggested activities.
- Access customized activity recommendations.

1.4.2 Organizer

- Publish and submit new event postings.
- Modify, update, call off, or mark listings as completed.
- Configure participant limits and registration closing times.
- Review current attendee registration lists.
- Examine essential signup and turnout metrics.

1.4.3 Administrator

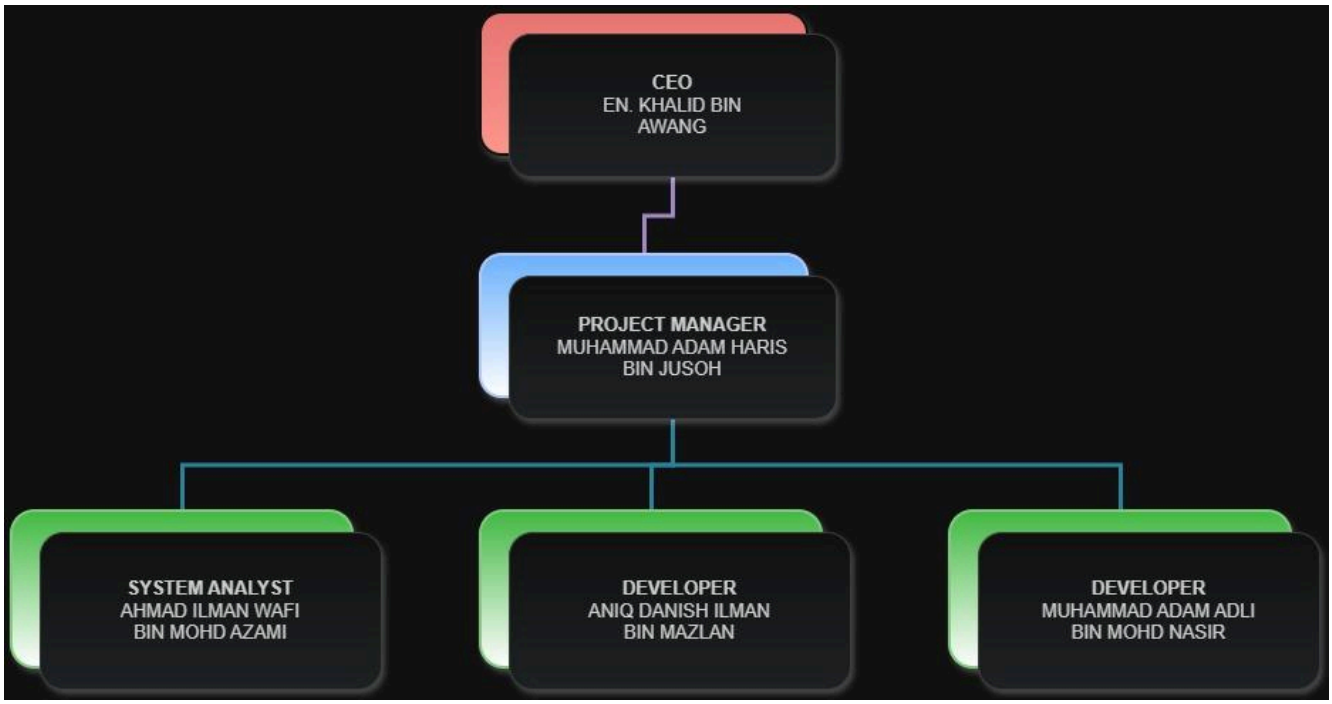
- Oversee participant accounts, event hosts, and recognized bodies.
- Authorize or decline pending event entries.
- Maintain and structure activity classifications.
- Control individual profile operational statuses.
- Track overall event trends and participant participation analytics.

1.4.4 System

- Maintain records of accounts, participants, organizers, organizations, listings, categories, registrations, bookmarks, interests, skills, and system alerts.
- Apply permission controls based on assigned roles.
- Track and update lifecycle statuses for events and enrollments.
- Produce tailored suggestions using gathered profile background and participation history.
- Dispatch notification messages regarding event updates.

1.5 Organization and Resources

1.5.1 Organization Structure



1.5.2 Personnel

Name	Role	Responsibilities
MUHAMMAD ADAM HARIS BIN JUSOH	Project Manager	● Lead and coordinate the project development team. ● Plan and monitor the overall project progress. ● Coordinate tasks and responsibilities among team members. ● Manage project schedules, milestones, and deliverables. ● Ensure the project meets its objectives and requirements. ● Monitor the team's progress and resolve project-related issues. ● Communicate with the lecturer/supervisor regarding project development.
AHMAD ILMAN WAFI BIN MOHD AZAMI	System Analyst	● Identify and analyze the requirements of the UniScout system.

		● Gather and document functional and non-functional requirements. ● Analyze the existing problems in the university event discovery and registration process. ● Design and document system processes and workflows. ● Prepare system specifications and requirements documentation. ● Ensure the proposed system meets user needs. ● Work closely with developers to clarify system requirements.
MUHAMMAD ADAM ADLI BIN MOHD NASIR	Developer	● Develop and implement the UniScout system. ● Design and develop the system's user interfaces. ● Implement system functions such as event browsing, keyword search, filtering, and interactive calendar. ● Develop participant profile management, interest tagging, and skill selection features. ● Implement personalized event recommendations based on participant profile and participation history. ● Develop notification alerts for reminders, registration deadlines, and event cancellations. ● Perform system testing and debugging. ● Maintain and improve the system based on project requirements.
ANIQ DANISH ILMAN BIN MAZLAN	Developer	● Develop and implement the UniScout system. ● Develop and integrate the database with the system. ● Implement user registration, login, and role-based access. ● Develop event registration and cancellation functions with capacity limits and waitlisting. ● Develop organizer functions for creating event listings, managing attendees, and attendance tracking. ● Develop admin functions for managing organizations, approving/rejecting events, categories, and analytics. ● Perform integration testing and identify system errors. ● Assist in debugging and improving system performance. ● Coordinate with the System Analyst and other developers throughout development.

1.5.3 Software Development Environment

- Hardware
    - Laptop
    - Smartphone
    - Printer
    - Stationery
    - A4 Paper
- Software
    - MySQL
    - MySQL Workbench
    - Google Chrome
    - Microsoft Word
    - Canva
    - Draw.io
    - Visual Studio Code
- Workspace
    - Web based UniScout: Event Management System

1.5.4 Software Development Library

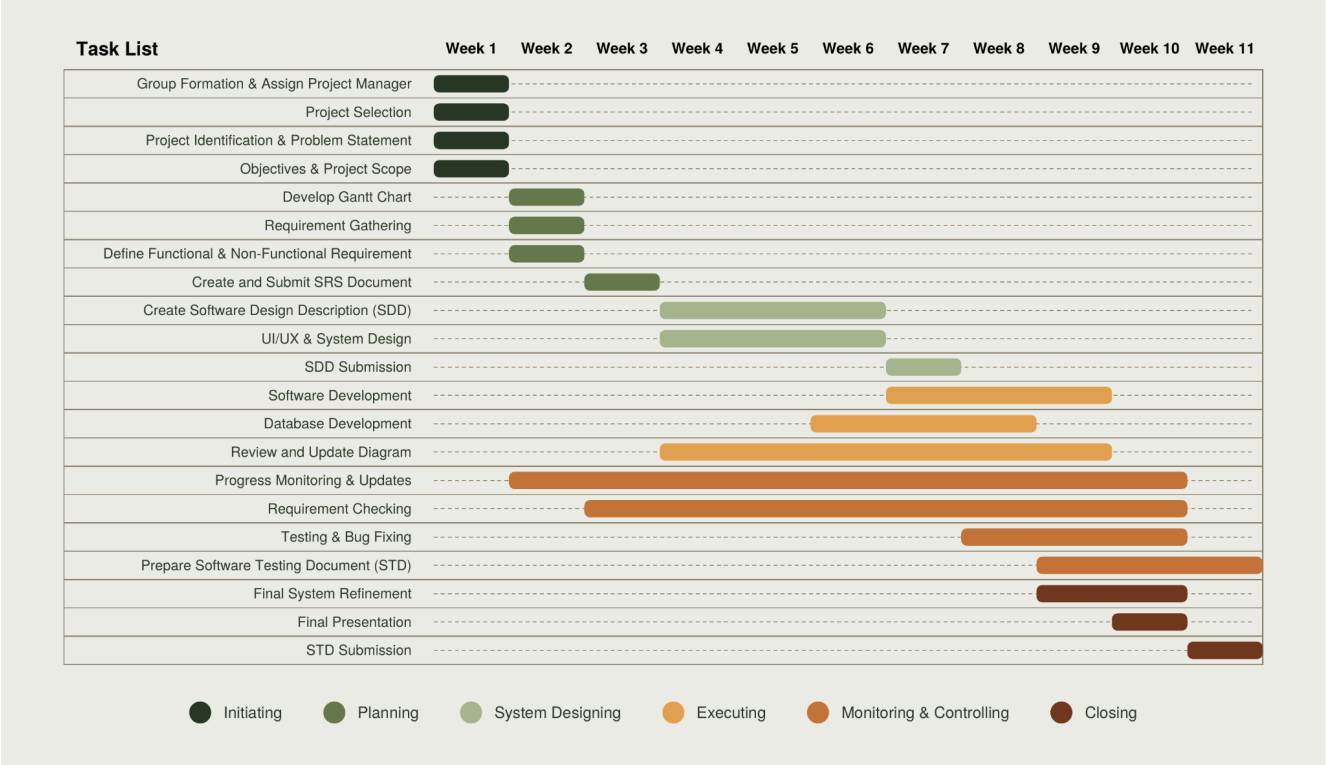
- Google Docs
    - For real-time collaboration where all team members can work on the same document simultaneously.
- WhatsApp
    - Main platform for quick communication, schedule meetings and information sharing.
- Github
    - Serves as the central code and asset version control repository, manages software source code, database migration scripts, and technical markdown files.

1.5.5 Configuration Identification

Element	Configuration
Functional Requirement	**FR-XXX-001** ● **FR**: Functional Requirement ● **XXX**: Represents the stakeholder (PARTICIPANT, ORGANIZER, ADMIN) ● **001**: Numbering in sequence according to FR ● FR-PARTICIPANT-001 until FR-PARTICIPANT-014 are used to identify all Functional Requirements for Participants ● FR-ORGANIZER-001 until FR-ORGANIZER-009 are used to identify all Functional Requirements for Organizers ● FR-ADMIN-001 until FR-ADMIN-009 are used to identify all Functional Requirements for Administrators
Non-Functional Requirement	**NFR-XXX-001** ● **NFR:** Non-Functional Requirement ● **XXX:** Represents the principal non-functional constraint ● **001:** Numbering in sequence according to NFR ● NFR-SEC-001 to NFR-SEC-004: Security requirements. ● NFR-REL-001 to NFR-REL-002: Reliability requirements. ● NFR-AVA-001 to NFR-AVA-002: Availability requirements. ● NFR-PER-001 to NFR-PER-003: Performance requirements. ● NFR-POR-001: Portability requirement. ● NFR-MAIN-001 to NFR-MAIN-003: Maintainability requirements. ● NFR-DATA-001 to NFR-DATA-003: Data Integrity requirements. ● NFR-USE-001 to NFR-USE-002: Usability requirements. ● NFR-SCA-001: Scalability requirement.

1.6 Schedule and Milestone

1.6.1 Gantt Chart



1.6.2 Milestone

Milestone	Date	Status	Responsible
Initiating ● Group Organization ● Topic discussion ● Problem Statement ● System Objective & Scope	Week 1	Completed	All members
Planning ● Gantt chart ● Requirement Gathering ● SRS Document	Week 2-3	Completed	All members
System Designing ● SDD ● UI/UX Design ● System Design	Week 4-7	Upcoming	All members
Database Development ● Database Design & Implementation	Week 6-8	Upcoming	Developers
System Development ● Participant Module ● Organizer Module ● Admin Module	Week 7-9	Upcoming	Developers
Monitoring & Controlling ● Progress Monitoring & Updates ● Requirement Checking	Week 8-10	Upcoming	Developers & System Analyst
Testing and Implementation ● System Testing ● Error Identification ● Bug Fixing	Week 7-10	Upcoming	All members
Closing ● Final Presentation ● STD Submission	Week 9-11	Upcoming	All members

CHAPTER 2: REQUIREMENT SPECIFICATIONS

2.1 Functional Requirements

The Participant shall be able to:

FR-PARTICIPANT-001 The system shall allow participants to register for an account.

FR-PARTICIPANT-002 The system shall allow participants to log in and log out.

FR-PARTICIPANT-003 The system shall allow participants to view and update profile information.

FR-PARTICIPANT-004 The system shall allow participants to browse published events.

FR-PARTICIPANT-005 The system shall allow participants to search, filter, and sort events.

FR-PARTICIPANT-006 The system shall display events in a calendar interface.

FR-PARTICIPANT-007 The system shall display event details and eligibility information.

FR-PARTICIPANT-008 The system shall allow participants to register when registration is open.

FR-PARTICIPANT-009 The system shall allow participants to cancel their event registration prior to the registration deadline.

FR-PARTICIPANT-010 The system shall allow participants to bookmark and remove bookmarked events.

FR-PARTICIPANT-011 The system shall allow participants to view registration status and history.

FR-PARTICIPANT-012 The system shall allow participants to view system notifications and reminders.

FR-PARTICIPANT-013 The system shall generate personalized event recommendations based on participant profile and participation history.

FR-PARTICIPANT-014 The system shall allow participants to maintain their interest tags and skill sets.

The Organizer shall be able to:

FR-ORGANIZER-001 The system shall allow organizers to log in and log out.

FR-ORGANIZER-002 The system shall allow organizers to create event listings.

FR-ORGANIZER-003 The system shall allow organizers to save drafts.

FR-ORGANIZER-004 The system shall allow organizers to submit listings for review.

FR-ORGANIZER-005 The system shall allow organizers to edit listings they manage.

FR-ORGANIZER-006 The system shall allow organizers to configure event dates, deadlines, capacity, mode, and location.

FR-ORGANIZER-007 The system shall allow organizers to cancel or complete events.

FR-ORGANIZER-008 The system shall allow organizers to view registrations for their events.

FR-ORGANIZER-009 The system shall allow organizers to view basic registration and participation metrics.

The Administrator shall be able to:

FR-ADMIN-001 The system shall allow administrators to log in and log out.

FR-ADMIN-002 The system shall allow administrators to manage user accounts.

FR-ADMIN-003 The system shall allow administrators to activate or suspend accounts.

FR-ADMIN-004 The system shall allow administrators to review submissions.

FR-ADMIN-005 The system shall allow administrators to approve or reject event submissions.

FR-ADMIN-006 The system shall allow administrators to manage organization records and verification.

FR-ADMIN-007 The system shall allow administrators to manage event categories.

FR-ADMIN-008 The system shall allow administrators to view event and registration records.

FR-ADMIN-009 The system shall allow administrators to review basic platform analytics.

2.2 Non-functional Requirements

2.2.1 Security

NFR-SEC-001 The system shall require authentication for protected functions.

NFR-SEC-002 The system shall store passwords using secure password hashing.

NFR-SEC-003 The system shall enforce role-based authorization.

NFR-SEC-004 The system shall validate inputs and protect personal and registration data.

2.2.2 Reliability

NFR-REL-001 The system shall preserve submitted records accurately and display errors when operations fail.

NFR-REL-002 The database shall be backed up regularly and be recoverable.

2.2.3 Availability

NFR-AVA-001 The system shall be available when users access published events, subject to planned maintenance.

NFR-AVA-002 Planned maintenance shall be communicated where feasible.

2.2.4 Performance

NFR-PER-001 Common pages and search/filter requests shall respond within five seconds under typical load.

NFR-PER-002 Login validation shall return a result within five seconds under typical load.

NFR-PER-003 Concurrent-user capacity shall be confirmed against the deployment environment.

2.2.5 Portability

NFR-POR-001 The web application shall work on current major browsers with an internet connection

2.2.6 Maintainability

NFR-MAIN-001 The application shall use modular components and consistent naming.

NFR-MAIN-002 The system shall maintain error logs for troubleshooting.

NFR-MAIN-003 Technical documentation and source-code backups shall be maintained.

2.2.7 Data Integrity

NFR-DATA-001 The system shall prevent duplicate active registrations for the same participant and event.

NFR-DATA-002 The system shall maintain valid relationships between related records.

NFR-DATA-003 The system shall validate event dates, deadlines, and capacity before accepting registrations.

2.2.8 Usability

NFR-USE-001 The interface shall make discovery, calendar navigation, and registration understandable.

NFR-USE-002 The system shall display clear feedback after user actions.

2.2.9 Scalability

NFR-SCA-001 The application and database shall support growth in users, events, registrations, and bookmarks.

CHAPTER 3: SOLUTION SPECIFICATION

3.1 Context Diagram

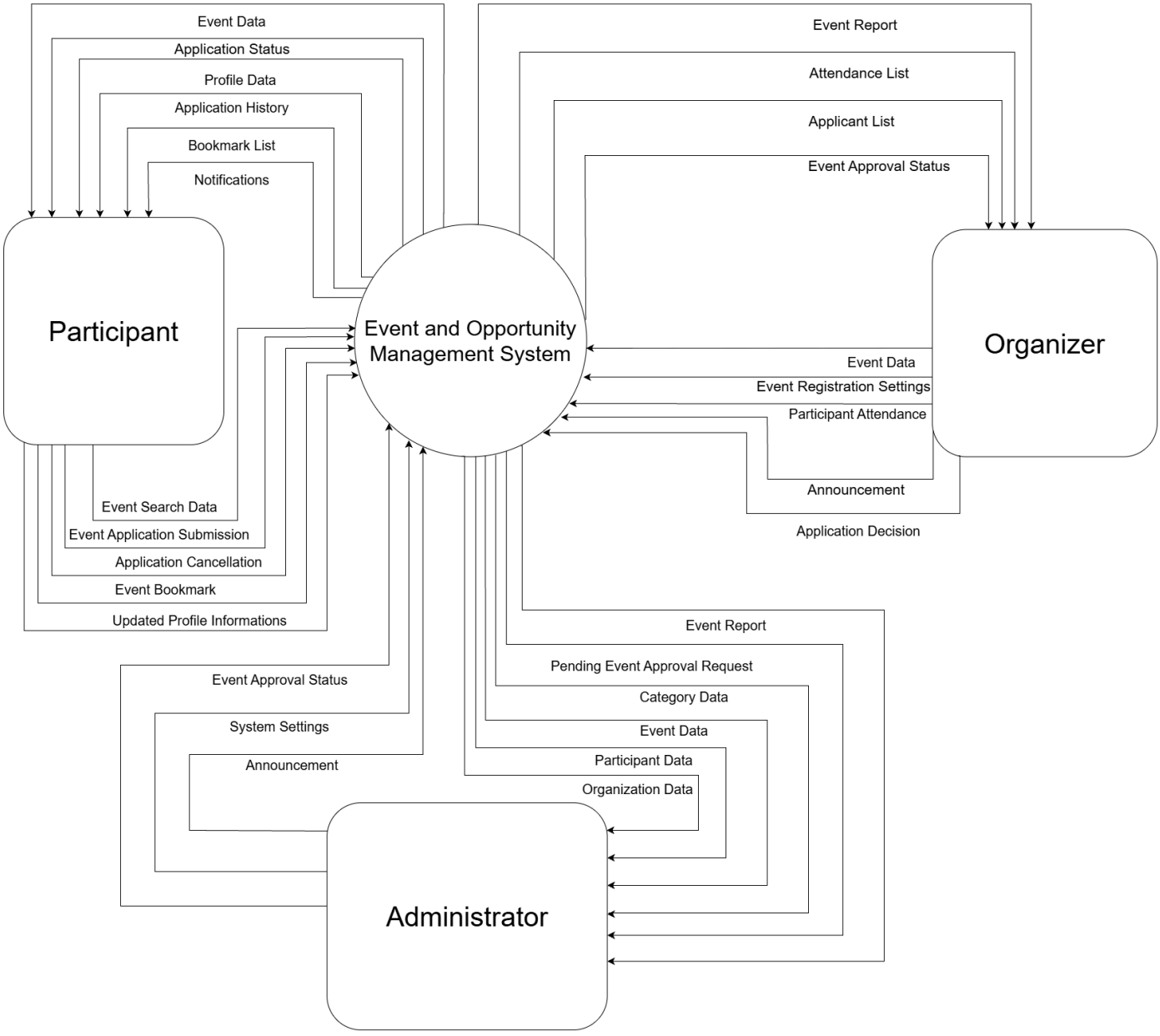


Figure 3.1.1: Context Diagram of Event and Opportunity Management System

Description of Context Diagram

The context diagram shows the Event and Opportunity Management System as a single process and its interactions with three external entities: **PARTICIPANT, ORGANIZER,** and **ADMINISTRATOR**. It defines the system boundary and the data that flows between the system and each entity, without showing the internal processes.

The **PARTICIPANT** sends event search data, event applications, application cancellations, event bookmarks, and updated profile information to the system. In return, the system provides event data, application status, profile data, application history, the bookmark list, and notifications.

The **ORGANIZER** submits event data, event registration settings, participant attendance, announcements, and application decisions. The system returns the event approval status, applicant list, attendance list, and event report.

The **ADMINISTRATOR** provides event approval status, system settings, and announcements to the system. In return, the system supplies pending event approval requests, category data, event data, participant data, organization data, and event reports for monitoring and management.

Overall, the diagram gives a high-level view of how the system supports event discovery and application for participants, event management for organizers, and approval and oversight for administrators.

3.2 Data Flow Diagram

3.2.1 Dfd Level 1

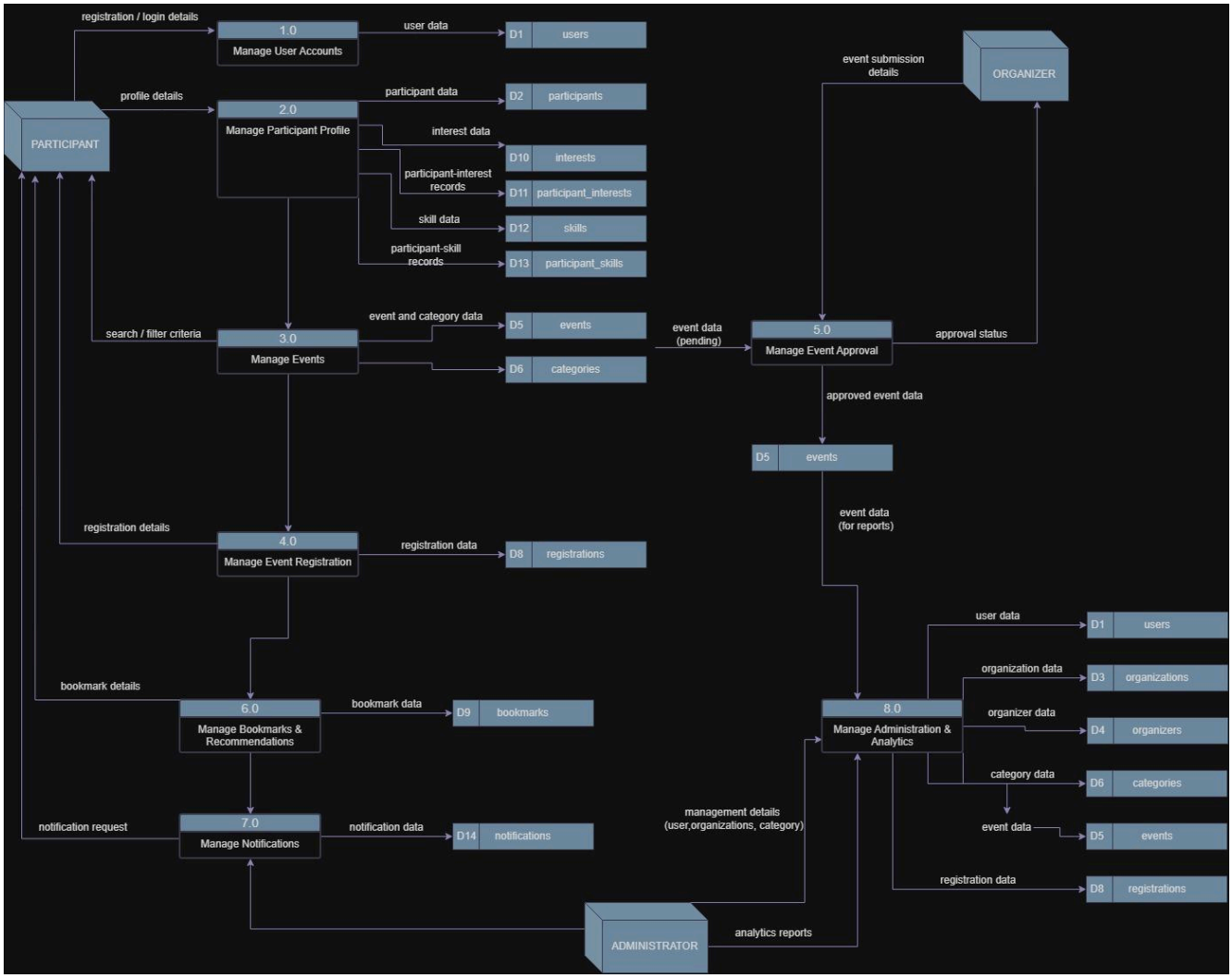


Figure 3.2.1

Description

The DFD consists of three external entities which are PARTICIPANT, ORGANIZER and ADMINISTRATOR. There are eight main processes in the system which is MANAGE USER ACCOUNTS, MANAGE PARTICIPANT PROFILE, MANAGE EVENTS, MANAGE EVENT REGISTRATION, MANAGE EVENT APPROVAL, MANAGE BOOKMARKS AND RECOMMENDATIONS, MANAGE NOTIFICATIONS and MANAGE ADMINISTRATOR AND ANALYTICS.

The system consists of fourteen data stores which are USERS, PARTICIPANT, ORGANIZATIONS, ORGANIZER, EVENTS, CATEGORIES, EVENT_CATEGORIES, REGISTRATION, BOOKMARKS, INTERESTS, PARTICIPANT_INTERESTS, SKILLS, PARTICIPANT_SKILLS and NOTIFICATIONS.

The data flow involved in the system are described as follows:

1. PARTICIPANT enters registration and details into the MANAGE USER ACCOUNTS (1.0) process. The process handles account information and stores the relevant user data in the USERS (D1) data store.
2. PARTICIPANT enters profile details into the MANAGE PARTICIPANT PROFILE (2.0) process. The process stores participant information in the PARTICIPANTS (D2) data store and manages interest and skill information using the INTERESTS (D10), PARTICIPANT_INTERESTS (D11), SKILLS (D12) and PARTICIPANT_SKILLS (D13) data stores.
3. PARTICIPANT enters search and filters criteria into the MANAGE EVENTS (3.0) process. The process retrieves event information from the EVENTS (D5) data store and provides available event details to the participant.
4. PARTICIPANT submits registration details to the MANAGE EVENT REGISTRATION (4.0) process. The process validates the registration information and stores the registration record in the REGISTRATIONS (D8) data store.
5. ORGANIZER submits event information through the MANAGE EVENT (3.0) process. The event details are stored in the EVENTS (D5) data store with the appropriate event status.
6. ORGANIZER submits an event for approval through the MANAGE EVENT APPROVAL (5.0) process. The process retrieves pending event information from the EVENTS (D5) data store for administrator review.
7. ADMINISTRATOR reviews the submitted event and provides an approval or rejection decision through the MANAGE EVENT APPROVAL (5.0) process. The process updates the event status in the EVENTS (D5) data store.
8. PARTICIPANT enters bookmark details into the MANAGE BOOKMARKS AND RECOMMENDATIONS (6.0) process. The process stores the bookmarked event information in the BOOKMARKS (D9) data store.
9. The MANAGE BOOKMARKS AND RECOMMENDATIONS (6.0) process retrieves participant interests, skills, participation history and event information to generate personalized recommendations for participants.
10. The MANAGE NOTIFICATIONS (7.0) process generates notifications related to event reminders, registration deadlines, cancellations and recommendations. The notification information is stored in the NOTIFICATION (D14) data store.
11. ADMINISTRATOR enters management details into the MANAGE ADMINISTRATOR AND ANALYTICS (8.0) PROCESS. The process manages information in the USERS (D1), ORGANIZATIONS (D3), ORGANIZERS (D4) and CATEGORIES (D6) data stores.
12. ADMINISTRATOR requests analytical reports through the MANAGE ADMINISTRATION AND ANALYTICS (D8) process. The process retrieves event and registration information from the EVENTS (D5) and REGISTRATIONS (D8) data stores to generate system reports.

3.2.2 Dfd Level 2

3.2.2.1 Manage User Accounts



Figure 3.2.2.1

Description

1. A USER inputs REGISTRATION DETAILS into the REGISTER ACCOUNT (1.1) process. The process stores NEW USER DATA in the USERS (D1) data store.
2. A USER submits a LOGOUT CREDENTIALS to the LOGIN (1.2) process. The process retrieves ACCOUNT DATA from the USERS (D1) data store to validate the login credentials.
3. A USER submits a LOGOUT REQUEST to the LOGOUT (1.3) process. The process ends the users session and provides a logout confirmation.
4. A USER submit a RESET PASSWORD REQUEST to the RESET PASSWORD (1.4) process. The process updates the NEW PASSWORD in the USERS (D1) data store.

3.2.2.2 Manage Participant Profile

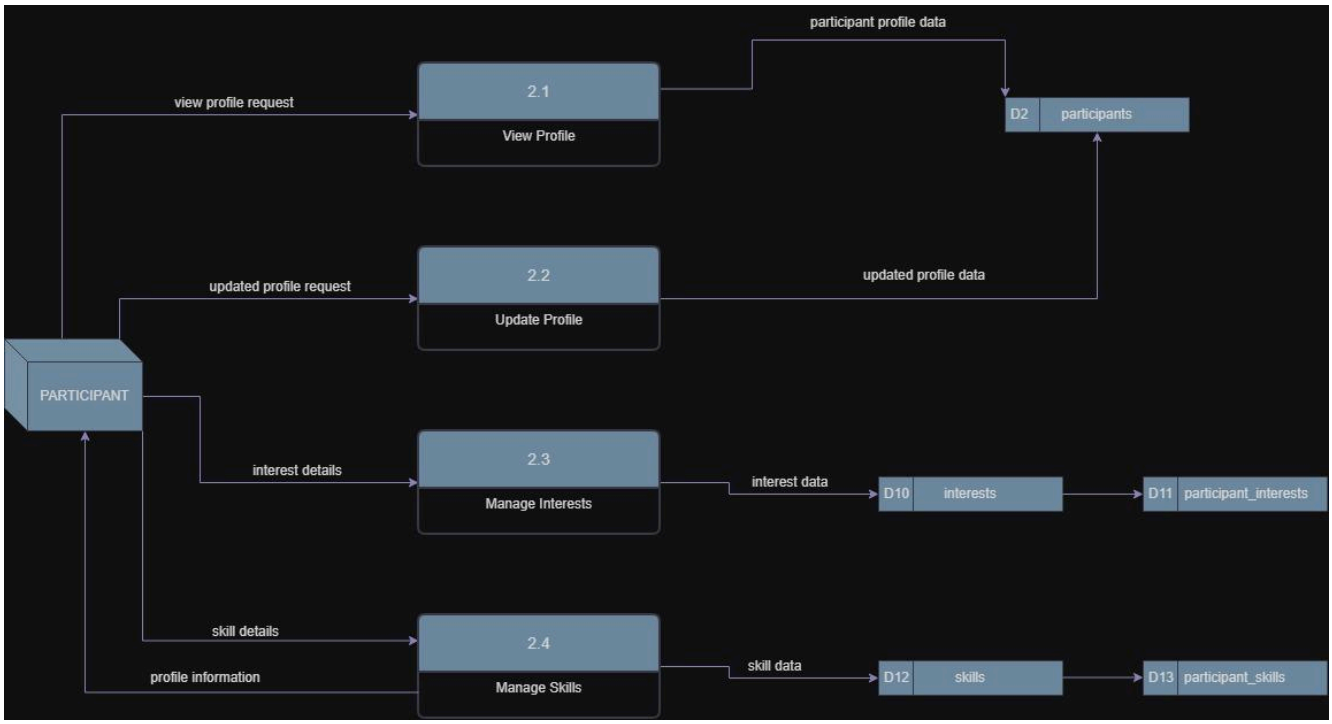


Figure 3.2.2.2

Description

1. A PARTICIPANT submits a VIEW PROFILE REQUEST to the VIEW PROFILE (2.1) process. The process retrieves PARTICIPANT PROFILE DATA from the PARTICIPANTS (D2) data store and displays the profile information.
2. A PARTICIPANT inputs UPDATED DETAILS into the UPDATE PROFILE (2.2) process. The process stores the UPDATED PROFILE DATA in the PARTICIPANTS (D2) data store.
3. A PARTICIPANT inputs INTEREST DETAILS into the MANAGE INTEREST (2.3) process. The process retrieves available interests from the INTERESTS (D10) data store and stores the selected interests in the PARTICIPANT_INTERESTS (D11) data store.
4. A PARTICIPANT inputs SKILL DETAILS into the MANAGE SKILLS (2.4) process. The process retrieves available skills from the SKILLS (D12) data store and stores the selected skills in the STUDENT_SKILLS (D13) data store.

3.2.2.3 Manage Events



Figure 3.2.2.3

Description

1. An ORGANIZER inputs EVENT CREATION DETAILS into th eCRETAE EVENT (3.1) process. The process stores NEW EVENT DATA in the EVENTS (D5) data store.
2. An ORGANIZER inputs EVENT UPDATE DETAILS into the EDIT EVENT (3.2) process. The process stores UPDATED EVENT DATA in the EVENTS (D5) data store.
3. A PARTICIPANT inputs SEARCH AND FILTER CRITERIA into the VIEW/SEARCH EVENTS (3.3) process. The process retrieves EVENT DATA from the EVENTS (D5) data store and displays the available event information.
4. An ORGANIZER submits EVENT CANCELLATION DETAILS to the CANCEL EVENT (3.4) process. The process updates the event status in the EVENTS (D5) data store.
5. An ADMINISTRATOR inputs CATEGORY MANAGEMENT DETAILS into the MANAGE EVENT CATEGORIES (3.5) process. The process stores CATEGORY DATA in the CATEGORIES (D6) data store and manages EVENT CATEGORY DATA in the EVENT_CATEGORIES (D7) data store.

3.2.2.4 Manage Event Registrations

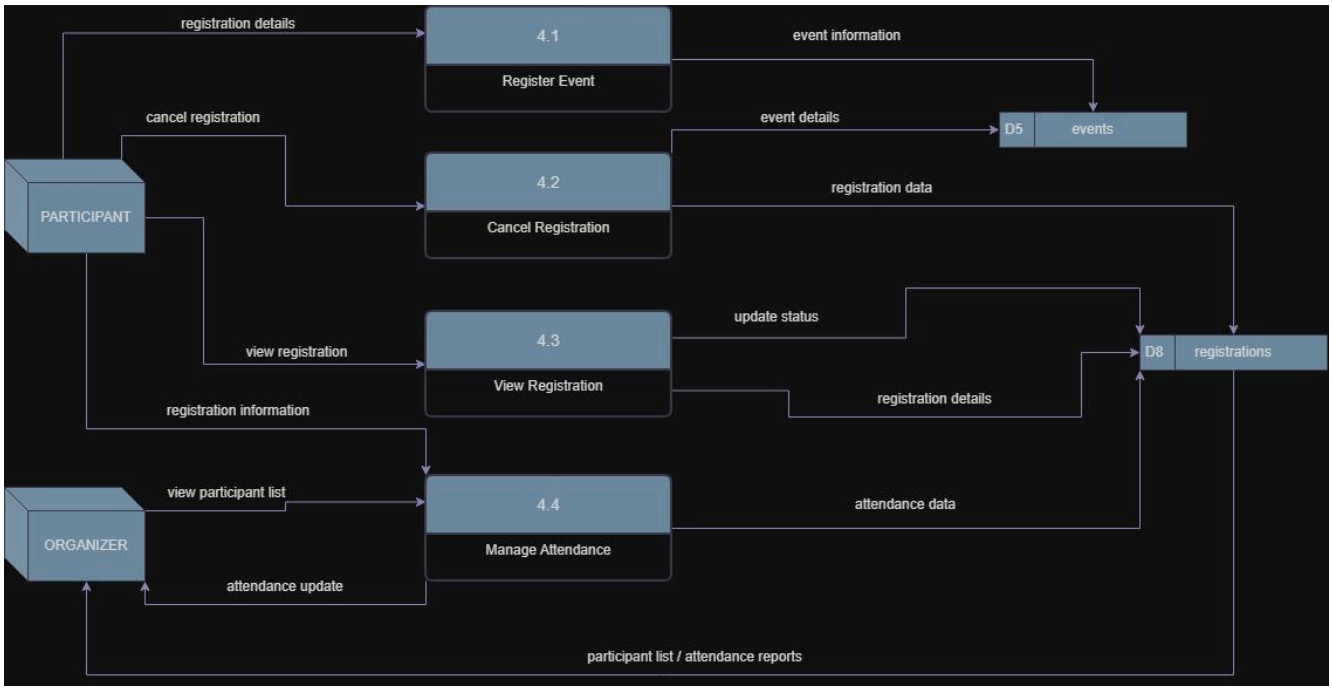


Figure 3.2.2.4

Description

1. A PARTICIPANT inputs REGISTRATION DETAILS into the REGISTER EVENT (4.1) process. The process retrieves EVENT INFORMATION from the EVENTS (D5) data store to validate the registration and stores the REGISTRATION DATA in the REGISTRATIONS (D8) data store.
2. A PARTICIPANT inputs CANCELLATION DETAILS into the CANCEL REGISTRATION (4.2) process. The process updates the REGISTRATION (D8) data store.
3. A PARTICIPANT submits a VIEW REGISTRATION REQUEST to the VIEW REGISTRATION (4.3) process. The process retrieves REGISTRATION DETAILS from the REGISTRATIONS (D8) data store and displays the registration information.
4. An ORGANIZER inputs ATTENDANCE UPDATE DETAILS into the MANAGE ATTENDANCE (4.4) process. The process stores ATTENDANCE DATA in the REGISTRATIONS (D8) data store and retrieves the participant list of the organizer.

3.2.2.5 Manage Event Approval

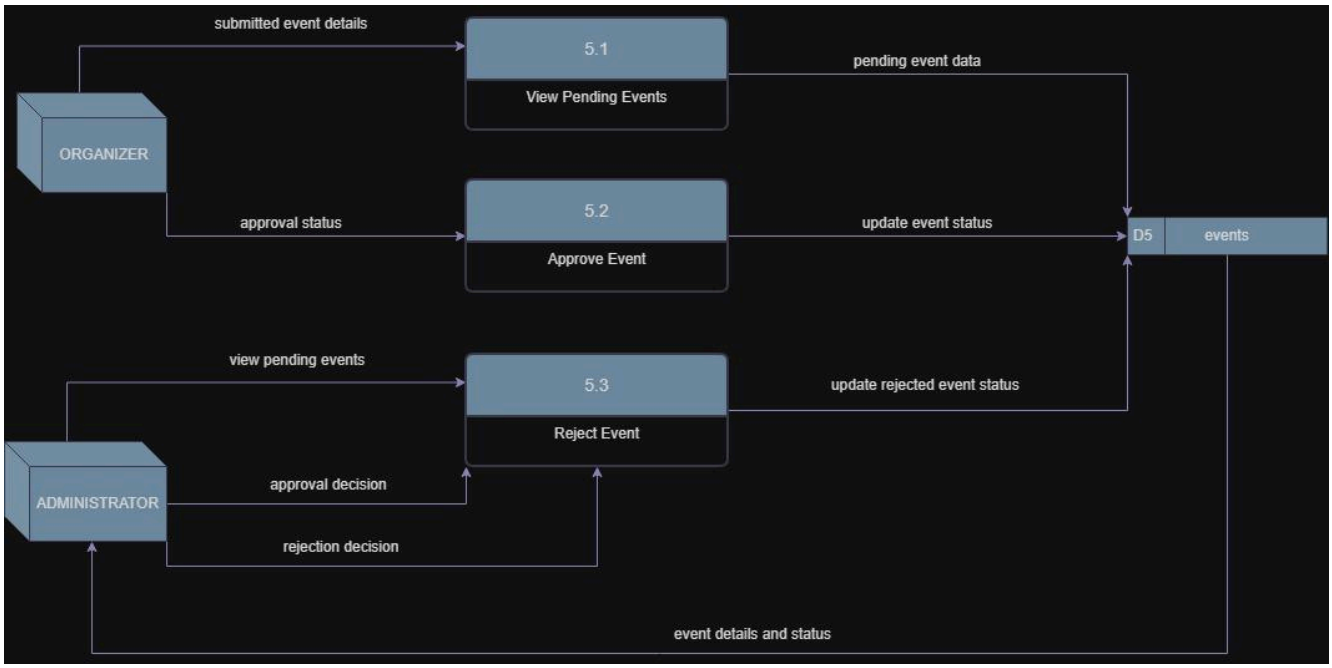


Figure 3.2.2.5

Description

1. An ORGANIZER submits EVENT DETAILS for approval. The VIEW PENDING EVENTS (5.1) process retrieves PENDING EVENT DATA from the EVENTS (D5) data store and displays the information to the ADMINISTRATOR.
2. An ADMINISTRATOR inputs an APPROVAL DECISION into the APPROVE EVENT (5.2) process. The process updates the EVENT STATUS to approved in the EVENTS (D5) data store.
3. An ADMINISTRATOR inputs a REJECTION DECISION into the REJECT EVENT (5.3) process. The process updates the EVENT STATUS to rejected in the EVENTS (D5) data store.
4. The system retrieves the updated EVENT STATUS from the EVENTS (D5) data store and sends the APPROVAL STATUS to the ORGANIZER.

3.2.2.6 Manage Bookmark And Recommendations

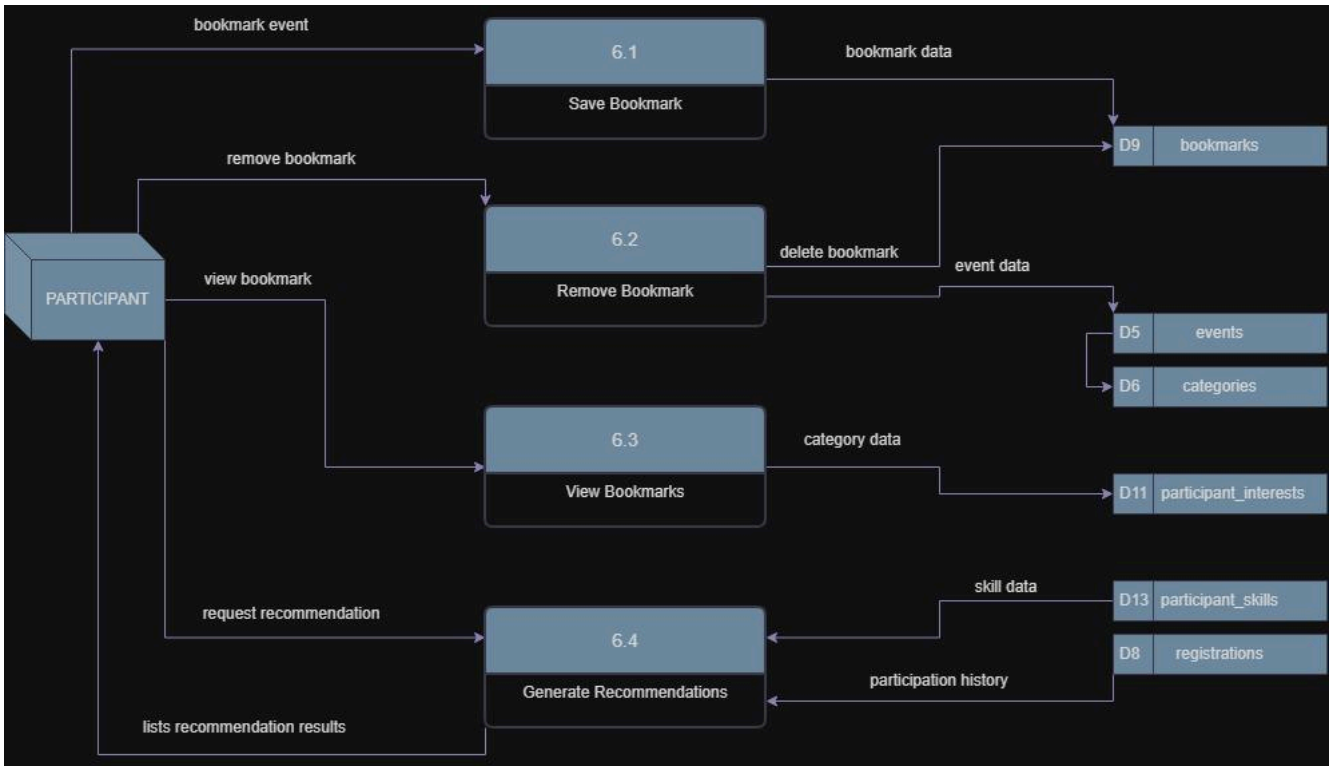


Figure 3.2.2.6

Description

1. A PARTICIPANT inputs BOOKMARK DETAILS into the SAVE BOOKMARK (6.1) process. The process stores BOOKMARK DATA in the BOOKMARKS (D9) data store.
2. A PARTICIPANT submits a REMOVE BOOKMARK REQUEST to the REMOVE BOOKMARK (6.2) process. The process removes the selected bookmark record from the BOOKMARK (D9) data store.
3. A PARTICIPANT submits a VIEW BOOKMARKS REQUEST to the VIEW BOOKMARKS (6.3) process. The process retrieves BOOKMARK DETAILS from the BOOKMARKS (D9) data store and displays the saved events.
4. A PARTICIPANT submits a RECOMMENDATION REQUEST to the GENERATE RECOMMENDATIONS (6.4) process. The process retrieves EVENT DATA from EVENTS (D5), CATEGORY DATA from CATEGORIES (D6), INTEREST DATA from STUDENT_INTERESTS (D11), SKILL DATA from STUDENT_SKILLS (D13) and PARTICIPATION HISTORY from REGISTRATION (D8).
5. The GENERATE RECOMMENDATIONS (6.4) process analyze the retrieved information and send PERSONALIZED RECOMMENDATION RESULTS to the STUDENT.

3.2.2.7 Manage Notifications

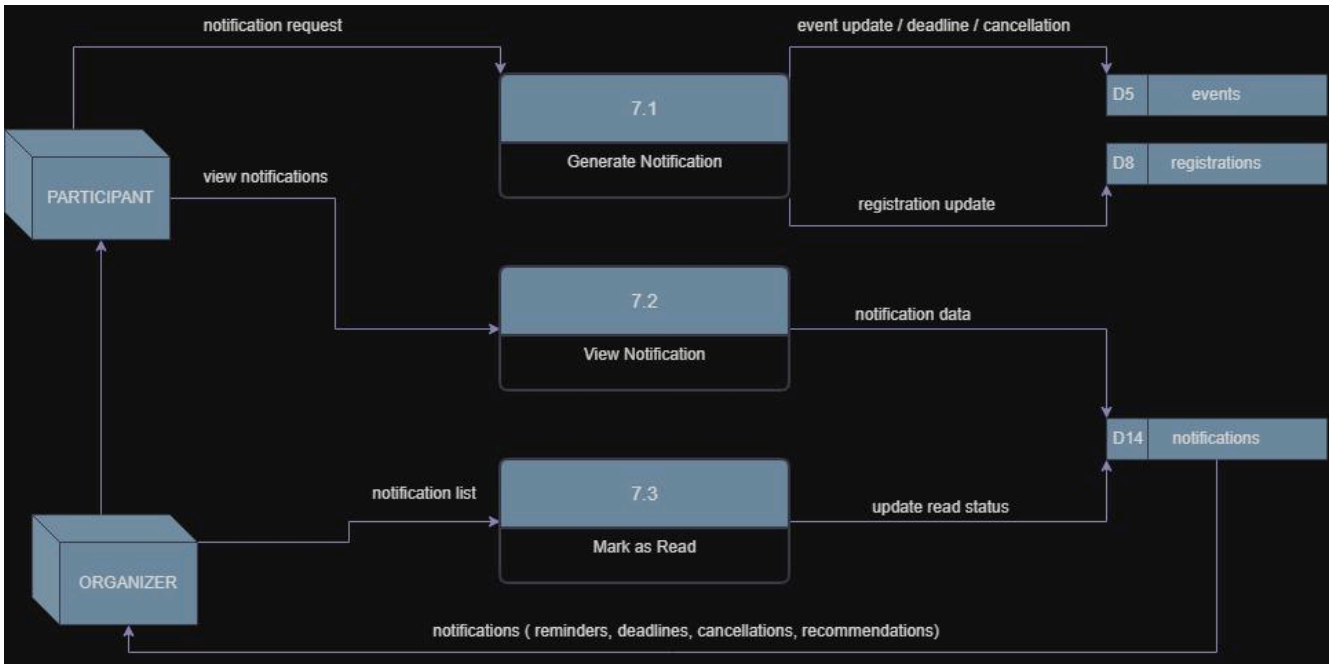


Figure 3.2.2.7

Description

1. The GENERATE NOTIFICATION (7.1) process retrieves EVENT INFORMATION from the EVENTS (D5) data store and REGISTRATION INFORMATION from the REGISTRATION (D8) data store. The process generates notifications and stores NOTIFICATION DATA in the NOTIFICATIONS (D14) data store.
2. A PARTICIPANT submits VIEW NOTIFICATION REQUEST to the VIEW NOTIFICATIONS (7.2) process. The process retrieves NOTIFICATION DATA from the NOTIFICATION (D14) data store and displays the notification information.
3. An ORGANIZER submits a VIEW NOTIFICATION REQUEST to the VIEW NOTIFICATIONS (7.2) process. The process retrieves relevant notification information from the NOTIFICATIONS (D14) data store.
4. A PARTICIPANT or ORGANIZER submits a MARK AS READ REQUEST to the MARK AS READ (7.3) process. The process updates the NOTIFICATION READ STATUS in the NOTIFICATIONS (D14) data store.

3.2.2.8 Manage Administration And Analytics

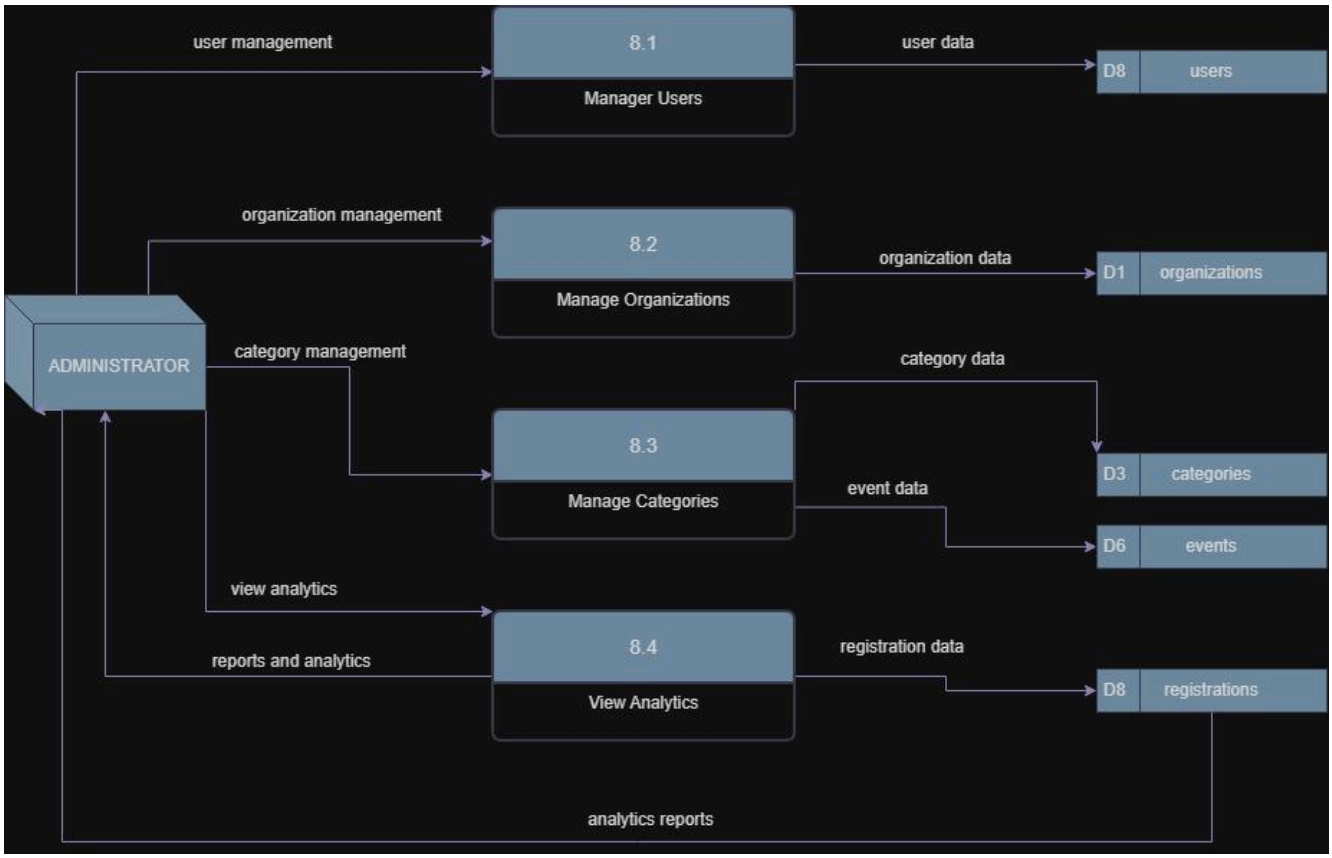


Figure 3.2.2.8

Description

1. An ADMINISTRATOR inputs USER MANAGEMENT DETAILS into the MANAGE USERS (8.1) process. The process stores UPDATED USER DATA in the USERS (D1) data store.
2. An ADMINISTRATOR inputs ORGANIZATION MANAGEMENT DETAILS into the MANAGE CATEGORIES (8.2) process. The process updates ORGANIZATION DATA in the ORGANIZATIONS (D3) data store and related organizer information in the ORGANIZERS (D4) data store.
3. An ADMINISTRATOR inputs CATEGORY MANAGEMENT DETAILS into the MANAGE CATEGORIES (8.3) process. The process stores UPDATED CATEGORY DATA in the CATEGORIES (D6) data store.
4. An ADMINISTRATOR submits an ANALYTICS REPORT REQUEST to the VIEW ANALYTICS (8.4) process. The process retrieves EVENT DATA from the EVENTS (D5) data store and REGISTRATION DATA from the REGISTRATION (D8) data store.
5. The VIEW ANALYTICS (8.4) process generates ANALYTICS REPORTS based on event and participation records and sends the reports to the ADMINISTRATOR.

3.2.3 Dfd Level 3

3.2.3.1 Validate And Process Registration

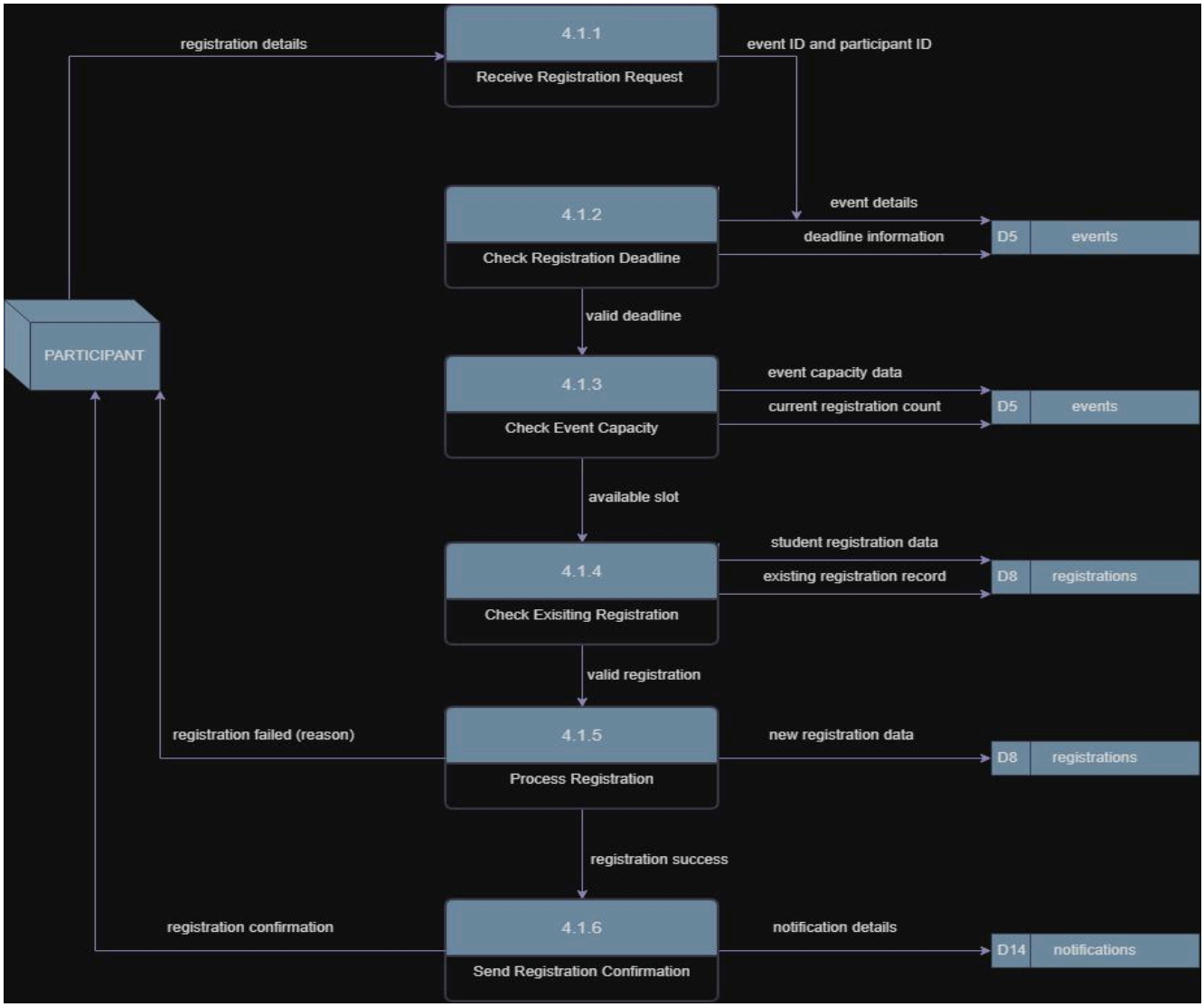


Figure 3.2.3.1

Description

1. A PARTICIPANT inputs REGISTRATION DETAILS into the RECEIVE REGISTRATION REQUEST (4.1.1) process. The process receives the EVENT ID and PARTICIPANT ID and sends the information to the CHECK REGISTRATION DEADLINE (4.1.2) process.
2. The CHECK REGISTRATION DEADLINE (4.1.2) process retrieves DEADLINE INFORMATION from the EVENTS (D5) data store. The process validates the registration deadline and send the VALID DEADLINE information to the CHECK EVENT CAPACITY (4.1.3) process.

3. The CHECK EVENT CAPACITY (4.1.3) process retrieves EVENT CAPACITY DATA from the EVENTS (D5) data store and CURRENT REGISTRATION RECORDS from the REGISTRATION (D8) data store. The process checks the available slots and sends the AVAILABLE SLOT information to the CHECK EXISTING REGISTRATION (4.1.4.) process.
4. The CHECK EXISTING REGISTRATIONS (4.1.4) process retrieves EXISTING REGISTRATION RECORDS from the REGISTRATION (D8) data store. The process checks for duplicate registrations and send the VALID REGISTRATION information to the PROCESS REGISTRATION (4.1.5) process.
5. The PROCESS REGISTRATION (4.1.5) process receives the validated registration information and stores NEW REGISTRATION DATA in the REGISTRATION (D8) data store. The process sends the REGISTRATION SUCCESS information to the SEND REGISTRATION CONFIRMATION (4.1.6) process.
6. The SEND REGISTRATION CONFIRMATION (4.1.6) process generates REGISTRATION CONFIRMATION and sends it to the PARTICIPANT. The process also stores NOTIFICATION DETAILS in the NOTIFICATIONS (D14) data store.
7. If the registration deadline has passed, the event capacity is full or a duplicate registration exists and the system send a REGISTRATION FAILED message to the PARTICIPANT with the relevant reason.

3.2.3.2 Generate Personalized Recommendations

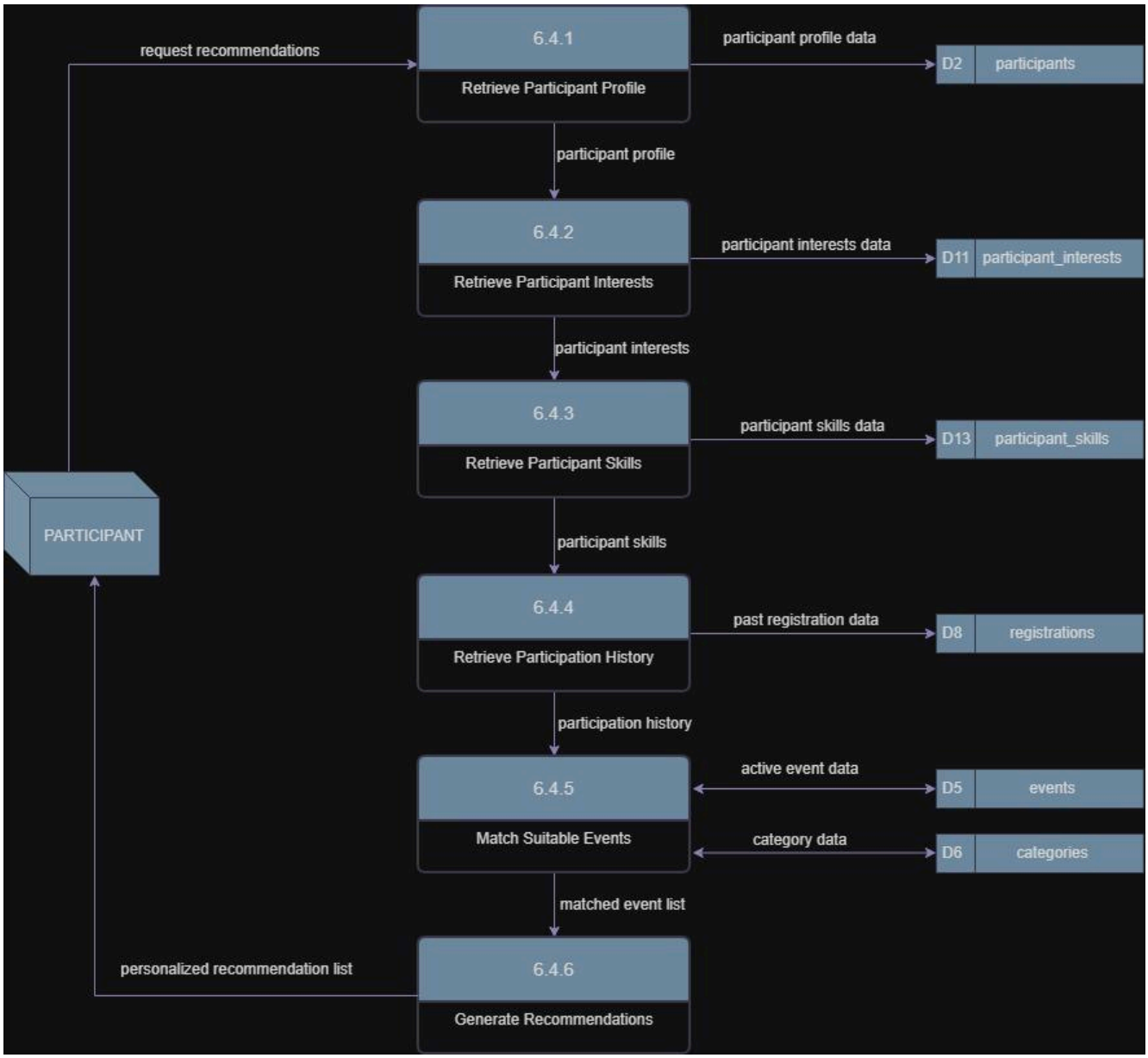


Figure 3.2.3.2

Description

1. A PARTICIPANT submits a RECOMMENDATION REQUEST to the RETRIEVE PARTICIPANT PROFILE (6.4.1) process. The process retrieves PARTICIPANT PROFILE DATA from the PARTICIPANTS (D2) data store and sends the profile information to the RETRIEVE PARTICIPANT INTERESTS (6.4.2) process.
2. The RETRIEVE PARTICIPANT INTERESTS (6.4.2) process retrieves PARTICIPANT INTEREST DATA from the PARTICIPANT_INTERESTS (D11) data store. The process sends the PARTICIPANT INTERESTS information to the RETRIEVE PARTICIPANT SKILLS (6.4.3) process.

3. The RETRIEVE PARTICIPANT SKILLS (6.4.3) process retrieves PARTICIPANT SKILL DATA from the PARTICIPANT_SKILLS (D13) data store. The process sends the PARTICIPANT SKILLS information to the RETRIEVE PARTICIPATION HISTORY (6.4.4) process.
4. The RETRIEVE PARTICIPATION HISTORY (6.4.4) process retrieves PAST REGISTRATION DATA from the REGISTRATION (D8) data store. The process sends the PARTICIPATION HISTORY information to the MATCH SUITABLE EVENTS (6.4.5) process.
5. The MATCH SUITABLE EVENTS process retrieves ACTIVE EVENT DATA from the EVENTS (D5) data store and CATEGORY DATA from the CATEGORIES (D6) data store. The process compares the available events with participant interests, skills and participation history.
6. The MATCH SUITABLE EVENTS (6.4.5) process sends the MATCHED EVENT LIST to the GENERATE RECOMMENDATIONS (6.4.6) process.
7. The GENERATE RECOMMENDATIONS (6.4.6) process generates a PERSONALIZED RECOMMENDATION LIST based on the matched events and send the recommendation results to the PARTICIPANT.

3.3 Capability Requirements

3.3.1 Participant

3.3.1.1 Participant Registration [FR-PARTICIPANT-001]

Field	Specification
Use Case	Participant Registration
Description	The participant registers for a new account to access protected features in the UniScout system.
Pre-condition	The user is an unregistered visitor on the UniScout registration page.
Basic Flow	1. The visitor clicks the “Register”/”Sign Up” button on the navigation bar. 2. The system displays the participant registration form. 3. The visitor enters their full name, participant ID, email address, password, and password confirmation. 4. The visitor clicks the “Submit” button. 5. The system validates the inputs, hashes the password, creates a new participant account in the database, and redirects the participant to the login page.
Alternative Flow	**1a. Redirect from Protected Action:** 1. The visitor browses published events and clicks “Register Now” or “Bookmark” without being logged in. 2. The system prompts the visitor to log in or sign up. 3. The visitor selects "Sign Up" and proceeds to Step 2 of the Basic Flow.
Exceptional Flow	**5a. Duplicate Email/ID or Invalid Input:** 1. If the email or participant ID is already registered, or if passwords do not match, the system displays a validation error message and prompts the visitor to correct the fields. **5b. Abandon Registration:** 1. The visitor may abandon the registration form at any time prior to submission.
Post-condition	A new participant account record is saved in the database with an active status, allowing the participant to log in.
Rules	1. Self-registration must be enabled. 2. Passwords must be stored using secure password hashing.

3.3.1.2 Participant Login and Logout [FR-PARTICIPANT-002]

Field	Specification
Use Case	Participant Login and Logout
Description	The participant logs into their account to access personalized features and logs out to end their authenticated session.
Pre-condition	The participant possesses a registered account in the UniScout system.
Basic Flow	1. The participant clicks “Login” on the navigation bar. 2. The participant enters their registered email/username and password. 3. The participant clicks the “Login” button. 4. The system verifies the credentials within five seconds and redirects the participant to the Participant Dashboard. 5. When finished, the participant clicks the “Logout” button. 6. The system terminates the active session and redirects the user to the public home page.
Alternative Flow	**1a. Credential Recovery:** 1. On the login page, the participant clicks "Forgot Password" to initiate credential recovery. **5a. Session Closure:** 1. The participant closes the browser window to end their interaction.
Exceptional Flow	**4a. Invalid Credentials or Suspended Account:** 1. If the email or password is incorrect, or if the account is suspended, the system rejects the login and displays an error message.
Post-condition	The participant is authenticated into their session (upon login) or their session is securely terminated (upon logout).
Rules	1. Authentication is required for all protected participant functions. 2. Login validation shall return a result within five seconds under typical load.

3.3.1.3 View and Update Participant Profile [FR-PARTICIPANT-003]

Field	Specification
Use Case	View and Update Participant Profile.
Description	The participant views and updates their personal and academic/affiliation profile details in UniScout.
Pre-condition	The participant is logged into the UniScout system.

Basic Flow	1. The participant navigates to the "My Profile" section. 2. The system retrieves and displays the participant's current profile data (participant ID, faculty/affiliation, program, academic year/level, and contact details). 3. The participant modifies the desired profile fields. 4. The participant clicks the "Save" button. 5. The system validates the inputs, updates the participant record in the database, and displays a confirmation message.
Alternative Flow	**3a. View Only / Cancel Edits:** 1. The participant reviews their profile information without saving changes or clicks "Cancel" to discard unsaved edits.
Exceptional Flow	**5a. Invalid or Empty Mandatory Fields:** 1. If required fields are cleared or formatted invalidly, the system blocks the update and displays a validation error message. **5b. Abandon Edit:** 1. The participant may abandon profile editing at any time before clicking “Save”.
Post-condition	The updated participant profile information is stored in the database and reflected across the system.
Rules	1. The participant must be logged in to edit their profile. 2. Profile updates immediately refresh the data used for eligibility checks and personalized recommendations.

3.3.1.4 Browse Published Events [FR-PARTICIPANT-004]

Field	Specification
Use Case	Browse Published Events.
Description	The participant explores the unified directory of approved, published campus events.
Pre-condition	The user accesses the UniScout Event Directory page.
Basic Flow	1. The participant selects the "Explore Events" / "Directory" section. 2. The system retrieves all events with a "Published" status from the database. 3. The system displays the event cards showing the title, organizer, category badge, event date, mode, and registration status within five seconds. 4. The participant scrolls through the catalogue or navigates between pages.
Alternative Flow	**2a. Switch Catalogue View:** 1. The participant toggles between grid view and list view, or switches to the Interactive Calendar.

Exceptional Flow	**2b. No Published Events Available:** 1. If no published events currently exist in the database, the system displays an empty-state message informing the participant that no active listings are available.
Post-condition	The system displays the list of currently published events to the participant.
Rules	1. Only events approved by an Administrator with a “Published” status are displayed in the public directory. 2. Common directory pages shall load within five seconds under typical load.

3.3.1.5 Search, Filter, and Sort Events [FR-PARTICIPANT-005]

Field	Specification
Use Case	Search, Filter, and Sort Events.
Description	The participant narrows down or reorders the event catalogue using keywords, domain/eligibility filters, and sorting criteria.
Pre-condition	The participant is on the Event Directory page.
Basic Flow	1. The participant enters a keyword in the search bar, selects filter parameters (category, mode, faculty/eligibility, or date range), or chooses a sort order (upcoming date, closing deadline, or newest). 2. The participant clicks “Search” / “Apply Filters”. 3. The system queries the database for published listings matching the specified criteria within five seconds. 4. The system refreshes the directory to display the matching, sorted events.
Alternative Flow	**2a. Clear / Reset Filters:** 1. The participant clicks "Clear Filters" to reset all search and filter parameters and restore the full list of published events.
Exceptional Flow	**3a. No Matching Results:** 1. If no events match the search keyword or filter combination, the system displays a “No matching events found” message and suggests clearing filters.
Post-condition	The directory displays the filtered and sorted list of published events matching the participant's criteria..
Rules	Search and filter requests shall respond within five seconds under typical load.

3.3.1.6 View Interactive Calendar [FR-PARTICIPANT-006]

Field	Specification
Use Case	View Interactive Calendar.
Description	The participant views published campus events and their registered events arranged in an interactive monthly, weekly, or daily calendar layout.
Pre-condition	The participant accesses the UniScout Calendar page.
Basic Flow	1. The participant clicks the "Calendar" tab in the navigation menu. 2. The system loads the default monthly calendar view populated with published events on their scheduled dates. 3. The participant switches between Monthly, Weekly, and Daily views or navigates to next/previous time periods. 4. The participant clicks an event entry on the calendar to view a quick summary and a link to its full details page.
Alternative Flow	**3a. Filter Calendar by Registered/Bookmarked Events:** 1. Authenticated participants toggle the calendar filter to display only their registered or bookmarked events.
Exceptional Flow	**2a. No Events in Selected Period:** 1. If no events fall within the selected month, week, or day, the calendar grid renders empty for those dates without system errors.
Post-condition	The participant views the chronological schedule of events in the selected calendar layout.
Rules	The calendar interface must make date navigation and schedule discovery clear and understandable.

3.3.1.7 View Event Details and Eligibility [FR-PARTICIPANT-007]

Field	Specification
Use Case	View Event Details and Eligibility.
Description	The participant views the complete information, schedule, venue/mode, remaining capacity, and eligibility criteria for a selected event.
Pre-condition	The event has a "Published" status and is selected from the directory, calendar, bookmark list, or recommendations.
Basic Flow	1. The participant clicks "View Details" on an event card or calendar item. 2. The system retrieves the event record, organizer details, category, schedule, capacity count, and eligibility criteria from the database. 3. The system displays the complete Event Details page, highlighting

	whether registration is open and whether the participant meets the eligibility requirements.
Alternative Flow	**3a. Proceed to Register on Bookmark:** 1. From the details page, the participant clicks "Register Now" or "Bookmark" .
Exceptional Flow	**2a. Event Removed or Cancelled:** 1. If the event was cancelled or unpublished before the page loads, the system displays an alert indicating the event is no longer active.
Post-condition	The full details and eligibility rules of the selected event are displayed on screen.
Rules	The system must clearly display the registration deadline, available capacity, and eligibility parameters prior to registration.

3.3.1.8 Register for Event [FR-PARTICIPANT-008]

Field	**Specification**
Use Case	Register for Event.
Description	The participant submits an enrollment registration for an open, published event.
Pre-condition	1. The participant is logged into the UniScout system. 2. The participant is viewing an event whose registration is currently open.
Basic Flow	1. The participant clicks the "Register Now" button on the Event Details page. 2. The system validates that the registration deadline has not passed, capacity is available, and the participant does not already have an active registration. 3. The system displays a confirmation prompt with the participant's details. 4. The participant clicks "Confirm". 5. The system saves the registration record in the database, updates the available slot count, and displays a confirmation alert.
Alternative Flow	**4a. Abort Confirmation:** 1. The participant clicks "Cancel" on the confirmation prompt, returning to the details page without registering.
Exceptional Flow	**2a. Capacity Full, Deadline Passed, or Duplicate Registration:** 1. the event is full, past its deadline, or already registered by the participant, the system blocks registration and displays an explanatory error message.
Post-condition	A new active registration record is stored in the database linking the participant to the event.
Rules	1. The system shall prevent duplicate active registrations for the same

	participant and event. 2. The system shall validate event dates, deadlines, and capacity before accepting registrations.

3.3.1.9 Cancel Event Registration [FR-PARTICIPANT-009]

Field	Specification
Use Case	Cancel Event Registration.
Description	The participant withdraws an existing active registration for an upcoming event where cancellation is permitted.
Pre-condition	The participant is logged in and holds an active registration for an upcoming event.
Basic Flow	1. The participant navigates to "My Registrations" (or the Event Details page). 2. The participant selects an active upcoming registration and clicks "Cancel Registration". 3. The system prompts the participant to confirm the withdrawal. 4. The participant clicks "Yes, Cancel". 5. The system updates the registration lifecycle status to "Cancelled" in the database, releases the seat back to the event's available capacity, and displays a confirmation message.
Alternative Flow	**4a. Keep Registration:** 1. The participant clicks "No" on the confirmation prompt, the system aborts the cancellation and keeps the registration active.
Exceptional Flow	**2a. Cancellation Deadline Passed or Event Completed:** 1. If the event has already occurred or the registration cutoff has passed, the system disables cancellation and informs the participant that withdrawal is no longer permitted.
Post-condition	The participant's registration status is updated to "Cancelled" in the database, and the event slot count is incremented.
Rules	Registration cancellation is only permitted prior to the event's registration cutoff date.

3.3.1.10 Bookmark and Remove Bookmarks [FR-PARTICIPANT-010]

Field	Specification
Use Case	Bookmark and Remove Bookmarks.
Description	The participant saves interesting event listings to their personal bookmark list for future reference or removes existing bookmarks.
Pre-condition	The participant is logged into the UniScout system.
Basic Flow	1. While browsing the directory or viewing an event's details page, the participant clicks the "Bookmark" (Save) icon. 2. The system creates a bookmark record linking the participant and the event in the database. 3. The system updates the icon state to "Bookmarked" and displays feedback. 4. The participant can view all saved items under the "My Bookmarks" section.
Alternative Flow	**1a. Remove Bookmark:** 1. The participant clicks the active "Bookmarked" icon on a saved listing (or clicks "Remove" inside the "My Bookmarks" page). 2. The system deletes the bookmark record from the database and updates the UI state.
Exceptional Flow	**1b. Unauthenticated User Clicks Bookmark:** 1. If an unauthenticated visitor clicks the bookmark icon, the system redirects them to log in first.
Post-condition	The event is added to or removed from the participant's saved bookmarks in the database.
Rules	A participant can bookmark an event only once, clicking an active bookmark toggles it off.

3.3.1.11 View Registration Status and History [FR-PARTICIPANT-011]

Field	Specification
Use Case	View Registration Status and History.
Description	The participant views their current registration statuses and past event participation history.
Pre-condition	The participant is logged into the UniScout system.
Basic Flow	1. The participant clicks the "My Registrations" section. 2. The system retrieves all registration records associated with the

	participant from the database. 3. The system displays the records categorized by lifecycle status (Registered/Upcoming, Attended/Completed, and Cancelled). 4. The participant selects a tab or clicks an entry to view the full event details.
Alternative Flow	**4a. Cancel an Upcoming Registration:** 1. From the Upcoming tab, the participant clicks “Cancel Registration” to trigger Cancel Event Registration.
Exceptional Flow	**2a. No Registration History:** 1. If the participant has never registered for any event, the system displays an empty-state message with a button to browse the directory.
Post-condition	The participant's current and historical registration records are displayed accurately on screen.
Rules	The participant must be logged in to access their registration records.

3.3.1.12 View Notifications [FR-PARTICIPANT-012]

Field	**Specification**
Use Case	View Notifications.
Description	The participant views automated alerts regarding upcoming event schedules, registration deadlines, event updates/cancellations, and suggested activities.
Pre-condition	The participant is logged into the UniScout system.
Basic Flow	1. The participant clicks the “Notifications” bell icon or menu tab. 2. The system retrieves all system alerts dispatched to the participant ordered from newest to oldest. 3. The participant clicks on a notification item (reminder, event update, or cancellation notice). 4. The system marks the notification as "Read" and navigates the participant to the associated event details page.
Alternative Flow	**3a. Mark All as Read:** 1. The participant clicks “Mark All as Read” to clear the unread notification badge counter.
Exceptional Flow	**2a. No Notifications:** 1. If the participant has no alerts, the system displays “You have no notifications at this time”.
Post-condition	The participant views their system notifications and the selected notification's status is updated to **"**Read**"** in the database.

Rules	The system automatically dispatches notifications when registered/bookmarked events approach their cutoff dates or when an organizer updates or cancels an event.

3.3.1.13 View Personalized Recommendations [FR-PARTICIPANT-013]

Field	Specification
Use Case	View Personalized Recommendations.
Description	The system generates and displays tailored event suggestions based on the participant's profile attributes, interest tags, skill sets, and prior participation history.
Pre-condition	The participant is logged into the UniScout system.
Basic Flow	1. The participant accesses the Participant Dashboard or "Recommended for You" section. 2. The system analyzes the participant's stored profile data (affiliation/faculty, program/level, interest tags, skills) and past registration history 3. The system matches these attributes against upcoming published events. 4. The system displays a ranked list of recommended events to the participant.
Alternative Flow	**2a. Update Interests/Skills to Refresh Recommendations:** 1. The participant clicks "Customize Interests & Skills" to refine their recommendation results.
Exceptional Flow	**2b. Incomplete Profile:** 1. If a newly registered participant has not yet selected interests, skills, or attended events, the system displays general featured/upcoming campus events and prompts the participant to complete their profile.
Post-condition	The participant is represented with personalized event listings tailored to their profile and activity history.
Rules	Only active, published events whose registration deadlines have not passed are included in recommendations.

3.3.1.14 Maintain Interests and Skills [FR-PARTICIPANT-014]

Field	Specification
Use Case	Maintain Interests and Skills.
Description	The participant selects, adds, or removes personal interest tags and skill sets on

	their profile to improve event matching and recommendations.
Pre-condition	The participant is logged in and accessing the Profile Management page.
Basic Flow	1. The participant navigates to the “Interests & Skills” section within “My Profile”. 2. The system displays the available interest categories and skill tags alongside the participant’s currently selected tags. 3. The participant selects new interest/skill tags or deselects existing ones. 4. The participant clicks “Save Preferences”. 5. The system updates the participant’s interest and skill associations in the database and displays a confirmation message.
Alternative Flow	**4a. Discard Changes:** 1. The participant clicks "Cancel" to revert any unsaved tag selections.
Exceptional Flow	**5a. Database Save Error:** 1. If the update fails due to a connection or server error, the system preserves the previous state and displays an error alert.
Post-condition	The participant’s updated interest and skill records are saved in the database and used by the recommendation engine..
Rules	Interest and skill associations must maintain valid foreign-key relationships with the master taxonomy records.

3.3.2 Organizer

3.3.2.1 Organizer Login and Logout [FR-ORGANIZER-001]

Field	Specification
Use Case	Organizer Login and Logout.
Description	The organizer logs into the UniScout portal to manage event listings and logs out when finished.
Pre-condition	The organizer has a registered Organizer account in the UniScout system.
Basic Flow	1. The organizer navigates to the Login page. 2. The organizer enters their registered email/username and password. 3. The organizer clicks “Login”. 4. The system validates the credentials within five seconds, verifies the Organizer role, and redirects the user to the Organizer Dashboard. 5. To end the session, the organizer clicks “Logout”. 6. The system clears the session and redirects to the login page.
Alternative Flow	**5a Close Browser:** 1. The Organizer closes the browser tab to exit the application.
Exceptional Flow	**4a. Invalid Credentials or Unverified/Suspended Account:** 1. If credentials are invalid or the organizer account is suspended by an Administrator, the system denies access and displays an error message.
Post-condition	The organizer is authenticated into the Organizer Dashboard (on login) or logged out of the session (on logout).
Rules	Role-based access control restricts organizer functions strictly to authenticated Organizer accounts.

3.3.2.2 Create Event Listing [FR-ORGANIZER-002]

Field	Specification
Use Case	Create Event Listing.
Description	The organizer initializes a new campus event or event listing with its core descriptive and classification details.
Pre-condition	The organizer is logged into an active Organizer account.
Basic Flow	1. From the Organizer Dashboard, the organizer clicks "Create Event". 2. The system displays the event creation form. 3. The organizer enters the event title, description, category, target eligibility

	criteria, relevant skill/interest tags, and uploads promotional posters/media if applicable. 4. The organizer proceeds to configure the schedule, capacity, and location parameters.
Alternative Flow	**4a. Save Immediately as Draft:** 1. The organizer clicks "Save Draft" to store partial listing details without publishing.
Exceptional Flow	**3a. Invalid Input Format:** **1.** If text inputs exceed length limits or contain invalid characters, the system highlights the field with an error message. **3b. Abandon Creation:** 1. The organizer clicks "Cancel" to discard the form and return to the dashboard.
Post-condition	The core descriptive attributes of the event listing are captured and ready for draft saving or review submission.
Rules	Only verified, active Organizers can create event listings.

3.3.2.3 Save Event Draft [FR-ORGANIZER-003]

Field	**Specification**
Use Case	Save Event Draft.
Description	The organizer saves an incomplete or unsubmitted event listing as a draft so it can be edited and submitted later.
Pre-condition	The organizer is logged in and is creating or editing an event listing.
Basic Flow	1. While filling out the event form, the organizer clicks the "Save as Draft" button. 2. The system validates basic data formatting. 3. The system saves the event record in the database with the lifecycle status "Draft". 4. The system displays a "Draft saved successfully" notification and lists the item under the organizer's "Drafts" tab.
Alternative Flow	**1a. Submit Directly for Review:** 1. Instead of saving a draft, the organizer clicks "Submit for Review" once all mandatory fields are complete.
Exceptional Flow	**2a. Missing Event Title:** 1. If the organizer attempts to save a draft without a minimum identifying title, the system prompts the organizer to enter a title before saving.
Post-condition	The event record is stored in the database with status "Draft" and remains hidden

	from participants and administrator review queues.
Rules	Draft listings are visible only to the managing Organizer until formally submitted for review.

3.3.2.4 Submit Event Listing for Review [FR-ORGANIZER-004]

Field	**Specification**
Use Case	Submit Event Listing for Review.
Description	The organizer submits a completed event listing to the Administrator for verification and publication approval.
Pre-condition	The organizer is logged in and has completed all mandatory fields and configurations for a new or draft event listing.
Basic Flow	1. The organizer reviews the completed event details and clicks "Submit for Review". 2. The system validates that all mandatory fields, dates, deadlines, capacity, and location details are complete and valid. 3. The system updates the event lifecycle status to "Pending Approval" in the database. 4. The system notifies the Administrator of the new submission and displays a confirmation message to the organizer
Alternative Flow	**1a. Submit from Saved Drafts List:** 1. The organizer opens an existing "Draft" listing from their dashboard, verifies the fields, and clicks "Submit for Review".
Exceptional Flow	**2a. Incomplete Mandatory Fields or Invalid Schedule:** 1. If any required field is missing or the registration deadline occurs after the event start date, the system blocks submission and highlights the errors.
Post-condition	The event status is set to "Pending Approval" in the database and appears in the Administrator's review queue.
Rules	Every event listing must be reviewed and approved by an Administrator before becoming visible to participants.

3.3.2.5 Edit Event Listing [FR-ORGANIZER-005]

Field	**Specification**
Use Case	Edit Event Listing.
Description	The organizer modifies the details or parameters of an existing event listing that they manage.

Pre-condition	The organizer is logged in and is the authorized creator/manager of the selected event listing.
Basic Flow	1. The organizer navigates to "My Events" on the Organizer Dashboard. 2. The organizer selects an event listing and clicks "Edit". 3. The system loads the existing event data into the edit form. 4. The organizer modifies the desired fields (e.g., description, venue, capacity, or deadline). 5. The organizer clicks "Save Changes". 6. The system validates the updated data, saves the changes in the database, and dispatches update notifications to registered participants if the listing was already published
Alternative Flow	**5a. Cancel Edits:** 1. The organizer clicks "Cancel" to discard modifications and return to the dashboard.
Exceptional Flow	**6a Capacity Reduced Below Current Registrations:** 1. If the organizer attempts to reduce the maximum capacity below the number of currently registered participants, the system rejects the change and displays a validation error.
Post-condition	The updated event details are saved in the database, and registered participants are notified of schedule/venue changes.
Rules	Organizers may only edit event listings that belong to their account/organization.

3.3.2.6 Configure Event Dates, Deadlines, Capacity, Mode, and Location [FR-ORGANIZER-006]

Field	**Specification**
Use Case	Configure Event Dates, Deadlines, Capacity, Mode, and Location.
Description	The organizer configures the operational logistics of an event, including start/end timestamps, registration closing deadline, maximum participant capacity, delivery mode, and physical venue or online link.
Pre-condition	The organizer is logged in and creating or editing an event listing.
Basic Flow	1. Within the event form, the organizer navigates to the "Schedule, Capacity & Logistics" section. 2. The organizer sets the event start date/time, event end date/time, and registration closing deadline. 3. The organizer enters the maximum participant capacity limit. 4. The organizer selects the delivery mode (Physical, Online, or Hybrid) and enters the corresponding venue location or meeting URL. 5. The system validates the logistical rules when the organizer saves or submits the form.

Alternative Flow	**3a. Update Deadline or Capacity on Active Listing:** 1. The organizer extends the registration deadline or increases seat capacity on an existing listing via the Edit screen.
Exceptional Flow	**5a. Invalid Chronological Order or Non-Positive Capacity:** 1. If the end date is earlier than the start date, the registration deadline is after the event end date, or the capacity is set to ≤ 0, the system displays an error message and prevents saving.
Post-condition	The event's date, deadline, capacity, mode, and location parameters are stored in the database and enforced during participant registration.
Rules	Registration deadlines must precede the event start/end time, and capacity must be a positive integer.

3.3.2.7 Cancel or Complete Event [FR-ORGANIZER-007]

Field	Specification
Use Case	Cancel or Complete Event.
Description	The organizer updates the lifecycle status of a managed event by either cancelling an upcoming event or marking a concluded event as completed.
Pre-condition	The organizer is logged in and manages a published event listing.
Basic Flow	1. The organizer navigates to "My Events" and selects a managed listing. 2. After the event concludes, the organizer clicks "Mark as Completed". 3. The system prompts the organizer for confirmation. 4. The organizer confirms the action. 5. The system updates the event lifecycle status to "Completed" in the database and updates participant registration histories
Alternative Flow	**2a. Cancel Upcoming Event:** 1. Prior to the event date, the organizer clicks "Cancel Event" and enters a cancellation reason. 2. Upon confirmation, the system updates the event status to "Cancelled", closes registration, and dispatches cancellation alerts to all registered participants.
Exceptional Flow	**2b. Invalid Status Transition:** 1. If an event is already marked "Cancelled" or "Completed", the system disables further status transition buttons.
Post-condition	The event's status is updated to "Completed" or "Cancelled" in the database, and affected participants are notified.
Rules	Cancelling a published event automatically triggers system notifications to all enrolled participants.

3.3.2.8 View Event Registrations [FR-ORGANIZER-008]

Field	Specification
Use Case	View Event Registrations.
Description	The organizer views the list of participants who have registered for their managed event listings.
Pre-condition	The organizer is logged in and selects one of their managed events.
Basic Flow	1. The organizer navigates to "My Events" and selects an event. 2. The organizer clicks the "View Registrations" / "Attendee List" tab. 3. The system retrieves all participant registration records linked to that event from the database. 4. The system displays a table of registered participants including participant name, ID, faculty/affiliation, registration date, and registration status.
Alternative Flow	**4a. Search or Filter Attendee List:** 1. The organizer searches for a specific participant by name/ID or filters the list by registration status (Registered vs. Cancelled).
Exceptional Flow	**3a. Zero Registrations:** 1. If no participants have registered for the event yet, the system displays "No registrations recorded yet".
Post-condition	The current list of registered participants for the selected event is displayed to the organizer.
Rules	Organizers can only view registration lists for events they manage.

3.3.2.9 View Basic Registration and Participation Metrics [FR-ORGANIZER-009]

Field	Specification
Use Case	View Basic Registration and Participation Metrics.
Description	The organizer reviews summary statistics and turnout metrics for their managed events, such as total sign-ups, remaining capacity, cancellation counts, and participation rates.
Pre-condition	The organizer is logged into the Organizer Dashboard.
Basic Flow	1. The organizer navigates to the "Event Metrics" / "Analytics" section on the Organizer Dashboard (or within a specific event). 2. The system aggregates registration data across the organizer's events from the database. 3. The system displays key metrics including total active registrations, seat utilization percentage (registered vs. capacity), cancelled registrations,

	and breakdown by participant affiliation/faculty.
Alternative Flow	**1a. Filter Metrics by Event:** 1. The organizer selects a specific event from a dropdown menu to view metrics isolated to that single event.
Exceptional Flow	**2a. No Published Events:** 1. If the organizer has no published or past events, the metrics dashboard displays zero values across all KPI cards.
Post-condition	The system displays accurate registration and participation metrics to the organizer.
Rules	Metrics are calculated dynamically from live registration.

3.3.3 Administrator

3.3.3.1 Administrator Login and Logout [FR-ADMIN-001]

Field	Specification
Use Case	Administrator Login and Logout.
Description	The administrator logs into the UniScout Admin Portal to perform system oversight and logs out to terminate the session.
Pre-condition	The administrator holds an authorized Administrator account in the database.
Basic Flow	1. The administrator accesses the Login page. 2. The administrator enters their username/email and password. 3. The administrator clicks "Login". 4. The system verifies the credentials within five seconds, confirms Administrator role privileges, and redirects to the Admin Dashboard. 5. When administrative tasks are complete, the administrator clicks "Logout". 6. The system terminates the session and returns to the login screen.
Alternative Flow	**5a. Close Browser Window:** 1. The administrator closes the browser window to exit the portal.
Exceptional Flow	**4a. Invalid Credentials or Unauthorized Role:** 1. If login credentials fail validation or belong to a non-admin role attempting to access an admin URL, the system blocks access and displays an authentication error.
Post-condition	The administrator is authenticated into the Admin Portal (upon login) or securely logged out (upon logout).
Rules	Only accounts with the Administrator role can access administrative functions.

3.3.3.2 Manage User Accounts [FR-ADMIN-002]

Field	Specification
Use Case	Manage User Accounts.
Description	The administrator views, searches, filters, and manages participant and organizer user accounts across the platform.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator clicks "User Management" in the Admin navigation menu.

	2. The system retrieves and displays a paginated list of all registered accounts (Participants, Organizers, and Administrators). 3. The administrator searches for a specific account by name, ID, or email, or filters by role (Participant / Organizer). 4. The administrator selects a user account to view or update its profile details and assigned role. 5. The system saves any administrative updates to the database.
Alternative Flow	**4a. Create Organizer/Admin Account:** 1. The administrator clicks "Add User" to provision a new Organizer or Administrator account directly.
Exceptional Flow	**3a. No Matching User Found:** 1. If the search query matches no user records, the system displays "No matching user accounts found".
Post-condition	The administrator views the user directory and any account updates are persisted in the database.
Rules	User management actions are restricted exclusively to Administrators.

3.3.3.3 Activate or Suspend User Accounts [FR-ADMIN-003]

Field	**Specification**
Use Case	Activate or Suspend User Accounts.
Description	The administrator controls the operational status of a participant or organizer account by activating or suspending their access to the platform.
Pre-condition	The administrator is logged in and viewing the User Management list.
Basic Flow	1. The administrator locates the target participant or organizer account in the User Management table. 2. The administrator clicks "Suspend" (for an active account) or "Activate" (for a suspended/inactive account). 3. The system displays a confirmation dialog asking the administrator to confirm the status change. 4. The administrator clicks "Confirm". 5. The system updates the account status (Active or Suspended) in the database and displays a success message.
Alternative Flow	**4a. Cancel Status Change:** 1. The administrator clicks "Cancel" on the confirmation dialog; the account status remains unchanged.
Exceptional Flow	**2a. Self-Suspension Prevention:** 1. If the administrator attempts to suspend their own currently logged-in Admin account, the system blocks the action and displays an error

	message.
Post-condition	The target user's operational status is updated in the database; suspended users are blocked from logging in.
Rules	Suspending an account immediately revokes its ability to authenticate or perform actions until reactivated by an Administrator.

3.3.3.4 Review Event Submissions [FR-ADMIN-004]

Field	**Specification**
Use Case	Review Event Submissions.
Description	The administrator inspects pending event listings submitted by organizers to verify their completeness, eligibility rules, and compliance before publication.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator clicks "Event Submissions" in the Admin Portal. 2. The system queries the database and displays all event listings with the status "Pending Approval". 3. The administrator clicks "Review Details" on a pending submission. 4. The system displays the full submission details, including organizer organization, category, description, dates, deadline, capacity, and location.
Alternative Flow	**2a. Filter Submissions by Status or Organization:** 1. The administrator filters the list to view previously Approved or Rejected submissions or filters by a specific organization.
Exceptional Flow	**2b. No Pending Submissions:** 1. If the review queue has no pending listings, the system displays "No pending event submissions to review".
Post-condition	The administrator views the complete details of the submitted event and is ready to approve or reject it.
Rules	Only submissions formally submitted by an Organizer appear in the pending review queue.

3.3.3.5 Approve or Reject Event Submissions [FR-ADMIN-005]

Field	**Specification**
Use Case	Approve or Reject Event Submissions.
Description	The administrator authorizes a reviewed event submission to publish it to participants or declines it with feedback for the organizer.

Pre-condition	The administrator is logged in and reviewing a "Pending Approval" event submission.
Basic Flow	1. On the submission review screen, the administrator verifies all details and clicks "Approve". 2. The system updates the event's lifecycle status to "Published" in the database. 3. The system makes the event immediately discoverable in the Participant Directory and Calendar 4. The system sends an approval notification to the submitting organizer and displays a success message
Alternative Flow	**1a. Reject Submission:** 1. The administrator clicks "Reject", enters a reason/feedback remark, and confirms. 2. The system updates the event status to "Rejected" in the database and notifies the organizer with the rejection remarks.
Exceptional Flow	**1b. Reject Without Reason:** 1. If the administrator clicks "Reject" without entering feedback remarks, the system prompts the administrator to provide a mandatory reason before saving.
Post-condition	The event status is updated to "Published" or "Rejected" in the database, and the organizer is notified.
Rules	Rejected events are not displayed in the participant catalogue or calendar.

3.3.3.6 Manage Organization Records and Verification [FR-ADMIN-006]

Field	**Specification**
Use Case	Manage Organization Records and Verification.
Description	The administrator creates, updates, and verifies campus organizations (faculties, departments, clubs, and student bodies) and links them to organizer accounts.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator navigates to the "Organizations" management section. 2. The system displays all registered campus organizations and their verification statuses. 3. The administrator clicks "Add Organization" (or selects an existing organization to edit/verify). 4. The administrator enters or updates the organization name, type (Faculty, Club, Department), contact info, and sets the verification status to "Verified". 5. The administrator clicks "Save". 6. The system stores the updated organization record in the database

Alternative Flow	**4a. Revoke Verification / Deactivate Organization:** 1. The administrator changes an organization's status to "Unverified" or "Inactive", preventing new event submissions under that entity.
Exceptional Flow	**5a. Duplicate Organization Name:** 1. If an organization with the exact same name already exists, the system displays a duplicate record error.
Post-condition	The organization record and its verification status are updated in the database.
Rules	Only verified organizations can have their event listings published on UniScout.

3.3.3.7 Manage Event Categories [FR-ADMIN-007]

Field	Specification
Use Case	Manage Event Categories.
Description	The administrator creates, edits, or deactivates activity classification categories (e.g., Academic, Workshop, Hackathon, Volunteering, Career, Sports) used across the platform.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator selects the "Categories" management menu. 2. The system retrieves and displays the list of existing event categories. 3. The administrator clicks "Add Category" (or "Edit" on an existing category). 4. The administrator inputs the category name, description, and active status, then clicks "Save". 5. The system validates the input and saves the category record in the database.
Alternative Flow	**3a. Deactivate / Delete Unused Category:** 1. The administrator deletes an unused category or marks a category as inactive so it cannot be selected for future listings.
Exceptional Flow	**3b. Delete Category Linked to Existing Events:** 1. If the administrator attempts to delete a category that is already assigned to existing event records, the system blocks deletion to preserve referential integrity and advises deactivating it instead. **4a. Duplicate Category Name:** 1. If the category name already exists, the system displays an error message
Post-condition	The master list of event categories is updated in the database and reflected in Organizer forms and Participant filters.
Rules	Deleting categories with active foreign-key dependencies is prohibited to maintain relational data integrity.

3.3.3.8 View Event and Registration Records [FR-ADMIN-008]

Field	Specification
Use Case	View Event and Registration Records.
Description	The administrator audits all campus-wide event listings and participant registration records across the entire UniScout platform.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator navigates to the "All Events & Registrations" audit section. 2. The system retrieves and displays master tables of all events and participant registrations across all organizations. 3. The administrator filters records by event status, category, organization, date range, or participant ID. 4. The administrator clicks on any event or registration record to inspect its complete audit trail and details.
Alternative Flow	**3a. Switch Between Event View and Registration View:** 1. The administrator toggles between the master Events table and the master Participant Registrations table.
Exceptional Flow	**3b. No Matching Records:** 1. If a search query or filter returns zero records, the system displays an empty-state message.
Post-condition	The system displays the requested event and registration records to the administrator.
Rules	Administrators have read access to all platform event and registration records for auditing and oversight.

3.3.3.9 Review Basic Platform Analytics [FR-ADMIN-009]

Field	Specification
Use Case	Review Basic Platform Analytics.
Description	The administrator reviews platform-wide usage analytics, overall event trends, category distribution, and participant engagement metrics.
Pre-condition	The administrator is logged into the Admin Portal.
Basic Flow	1. The administrator navigates to the "Platform Analytics" dashboard. 2. The system queries the database to aggregate platform-wide statistics. 3. The system displays summary KPI cards and charts showing total active participants, verified organizations, total published events, overall

	registration volume, most popular event categories, and monthly participation trends
Alternative Flow	**3a. Filter Analytics by Time Period or Faculty/Category:** 1. The administrator selects a specific semester/date range or category filter to dynamically recalculate the analytics view.
Exceptional Flow	**2a. Insufficient Historical Data:** 1. If the platform has no recorded events or registrations within the selected date range, the charts display zero-state placeholders without throwing errors.
Post-condition	The administrator views up-to-date platform analytics and participation trends on screen.
Rules	Analytics queries must respond within five seconds under typical load.

3.4 Database

3.4.1 Entity Relationship Diagram

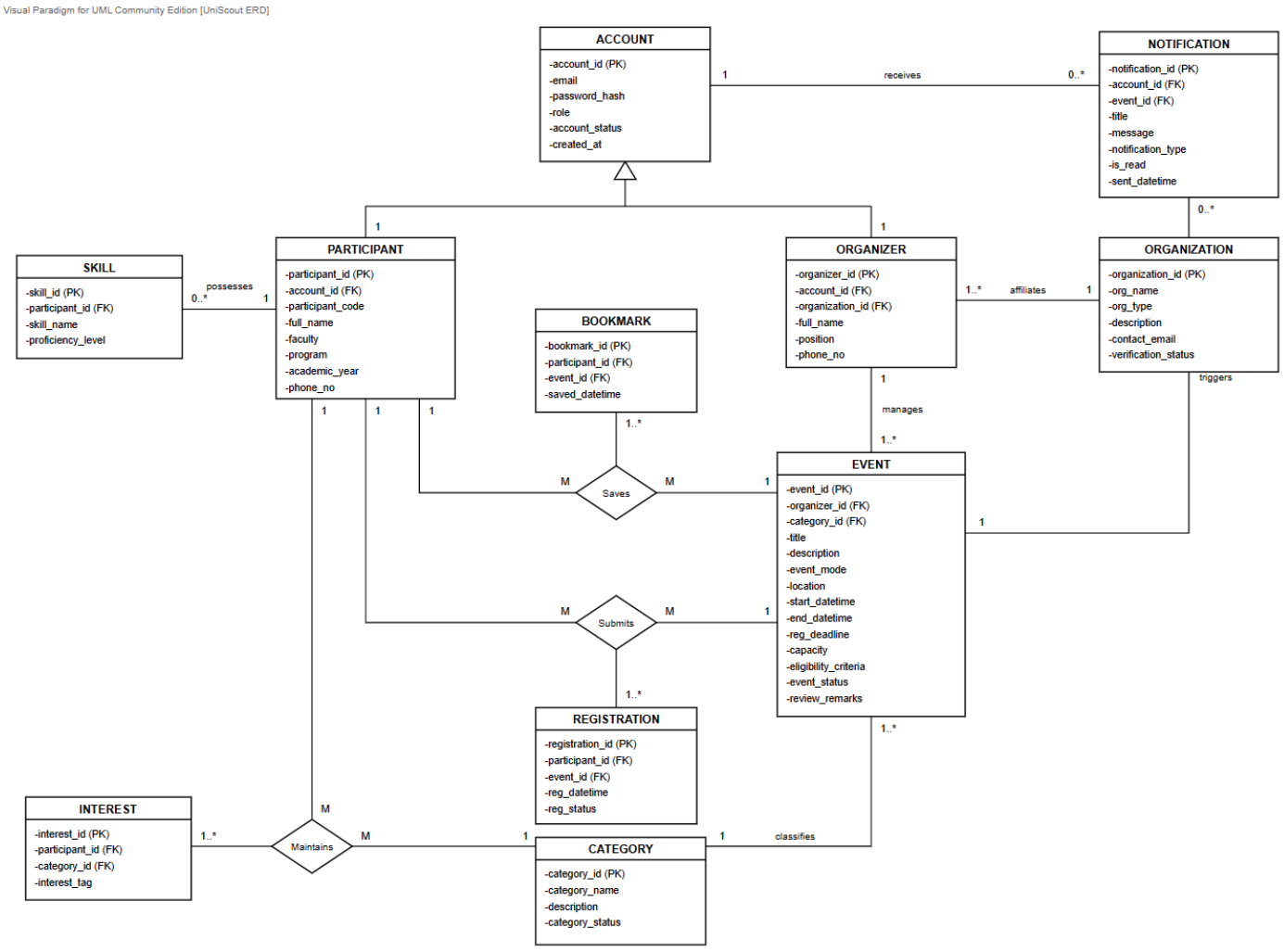


Figure 3.4.1

Figure 3.4.1 Entity Relationship Diagram

An Entity-Relationship Diagram (ERD) is use to illustrates the system's entities, attributes, and the relationships between entities. The Entity-Relationship Diagram composed of three main components which is identifying and defining the entities, determining the entities interactions, and specifying the cardinality of each relationship. As specified in system scope, the UniScout database eleven core entities: ACCOUNT, PARTICIPANT, ORGANIZER, ORGANIZATION, CATEGORY, EVENT, REGISTRATION, BOOKMARK, INTEREST, SKILL and NOTIFICATION.

Interaction of Account with Participant and Organizer

- Relationship: One-to-One (1: 1) specialization relationship.
- Description: The ACCOUNT entity stores centralized authentication credentials (email, password_hash), role assignments (PARTICIPANT, ORGANIZER, ADMIN), and operational status (account_status) for all system actors. An ACCOUNT with the Participant role maps to one PARTICIPANT profile record, while an ACCOUNT with the Organizer role maps to one ORGANIZER profile record. Administrators authenticate directly via the ACCOUNT entity to perform platform oversight.
- Keys: In ACCOUNT: account_id (PK).
  In PARTICIPANT: participant_id (PK), account_id (FK).
  In ORGANIZER: organizer_id (PK), account_id (FK).

Interaction of Organization and Organizer

- Relationship: One-to-Many (1:M) relationship.
- Description: One verified ORGANIZATION such as faculty, department, or club can be represented by one or many ORGANIZER accounts, while each ORGANIZER is affiliated with one specific ORGANIZATION.
- Keys: In ORGANIZATION:organization_id (PK).
  In ORGANIZER: organizer_id (PK), organization_id (FK)

Interaction of Organizer and Event

- Relationship: One-to-Many (1:M) relationship.
- Description: One ORGANIZER can create and manage one or many EVENT listings, while each EVENT listing is created and managed by one ORGANIZER.
- Keys: In ORGANIZER: organizer_id (PK)
  In EVENT: event_id (PK), organizer_id (FK).

Interaction of Category and Event

- Relationship: One-to-Many (1:M) relationship.
- Description: One CATEGORY such as Workshop or Hackathon can classify one or many EVENT listings, while each EVENT belongs to one primary CATEGORY.
- Keys: In CATEGORY: category_id (PK).
  In EVENT: event_id (PK), category_id (FK).

Interaction of Participant and Event (via Registration)

- Relationship: Many-to-Many (M:N) relationship resolved via the REGISTRATION associative entity.
- Description: A PARTICIPANT can register for one or many EVENT listings, and an EVENT can have one or many registered PARTICIPANT records. To resolve this Many-to-Many relationship, the associative entity REGISTRATION is created to track which participant

enrolled in which event, along with the registration timestamp and lifecycle status (Registered, Cancelled, Attended).

- Keys: In PARTICIPANT: participant_id (PK).
  In EVENT: event_id (PK).
  In REGISTRATION: registration_id (PK), participant_id (FK), event_id (FK).

Interaction of Participant and Event (via Bookmark)

- Relationship: Many-to-Many (M:N) relationship resolved via BOOKMARK associative entity.
- Description: A PARTICIPANT can bookmark one or many EVENT listings for future reference, and an EVENT can be bookmarked by one or many PARTICIPANT users. The associative entity BOOKMARK links the participant and the saved event.
- Keys: In BOOKMARK: bookmark_id (PK), participant_id (FK), event_id (FK).

Interaction of Participant with Interest and Skill

- Relationship: One-to-Many (1:M) relationships.
- Description: One PARTICIPANT can maintain multiple INTEREST tags (linked to event categories) and multiple SKILL records on their profile to power the personalized recommendation engine, while each individual INTEREST or SKILL entry belongs to one PARTICIPANT.
- Keys: In PARTICIPANT: participant_id (PK)
  In INTEREST: interest_id (PK), participant_id (FK), category_id (FK).
  In SKILL: skill_id (PK), participant_id (FK).

Interaction of Account and Event with Notification

- Relationship: One-to-Many (1:M) relationships.
- Description: One user ACCOUNT can receive one or many system NOTIFICATION alerts (reminders, deadline cutoffs, event cancellations, or approval alerts), and one EVENT can trigger one or many NOTIFICATION dispatches.
- Keys: In NOTIFICATION: notification_id (PK), account_id (FK), event_id (FK).

3.4.2 Data Dictionary

1.TABLE ACCOUNT

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	account_id	'account_id' represents the unique account identifier that is generated in auto-increment each time a user registers into the system. e.g 1	INT(10)	PRIMARY	10	0	NO
2	email	'email' represents the login email of the registered participant, organizer, or admin of the system. It will be used to verify their authorization to the system. e.g student@uniscout.edu.my	VARCHAR(100)	UNIQUE	100	NULL	NO
3	password_hash	'password_hash' represents the key to allow an authorized user or admin to login into the system. Password saved into Table Account is in encrypted/hashed form. e.g abc123 hashed	VARCHAR(255)	-	255	NULL	NO

4	role	'role' represents the user role in the system to restrict functions according to role ('participant', 'organizer', or 'admin'). e.g student	VARCHAR(20)	-	20	student	NO
5	account_status	'account_status' represents the status of the user account. 'A' indicates Active, 'S' indicates Suspended, and 'I' indicates Inactive. e.g A	CHAR(1)	-	1	A	NO
6	created_at	'created_at' represents the date and time when the account was created. It will be automatically generated in the system. e.g 2026-10-14 09:00:00	DATETIME	-	-	0000-00-00 00:00:00	NO

2.TABLE PARTICIPANT

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	participant_id	'participant_id' represents the unique student participant id that is generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	account_id	'account_id' represents the account_id from Table Account, which is the foreign key used to link the	INT(10)	FOREIGN	10	NULL	NO

		participant to their login account. e.g 1					
3	**participant_code**	**'participant_code' represents the unique university student matric/identification number. e.g 083370**	**VARCHAR(20)**	**UNIQUE**	**20**	**NULL**	**NO**
4	**full_name**	**'full_name' represents the full name of the registered student participant. e.g Muhammad Adam Haris bin Jusoh**	**VARCHAR(100)**	**-**	**100**	**NULL**	**NO**
5	**faculty**	**'faculty' represents the faculty name of the student participant. e.g Faculty of Informatics and Computing**	**VARCHAR(100)**	**-**	**100**	**NULL**	**YES**
6	**program**	**'program' represents the academic programme enrolled by the student. e.g Bachelor of Computer Science**	**VARCHAR(100)**	**-**	**100**	**NULL**	**YES**
7	**academic_year**	**'academic_year' represents the current year of study of the student. e.g 3**	**INT(2)**	**-**	**2**	**1**	**YES**

8	phone_no	'phone_no' represents the phone number or mobile number of the student participant. e.g 012-1234567	VARCHAR(15)	-	15	NULL	YES

3.TABLE ORGANIZATION

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	organization_id	'organization_id' represents the unique organization id that is generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	org_name	'org_name' represents the official name of the university organization. e.g Computer Science Society	VARCHAR(100)	-	100	NULL	NO
3	org_type	'org_type' represents the type of organization ('Faculty', 'Club', 'Department', or 'Society'). e.g Society	VARCHAR(50)	-	50	NULL	NO
4	description	'description' represents the background information and description of the organization.	VARCHAR(255)	-	255	NULL	YES

5	contact_email	'contact_email' represents the official contact email address of the organization. e.g css@uniscout.edu.my	VARCHAR(100)	-	100	NULL	NO
6	verification_status	'verification_status' represents the verification status of the organization. 'P' indicates Pending approval from admin, 'A' indicates Approved/Active, 'S' indicates Suspended, and 'R' indicates Rejected. e.g P	CHAR(1)	-	1	P	NO

4.TABLE ORGANIZER

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	organizer_id	'organizer_id' represents the unique organizer id that is generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	account_id	'account_id' represents the account_id from Table Account, which is the foreign key. e.g 2	INT(10)	FOREIGN	10	NULL	NO

3	**organization_id**	**'organization_id' represents the organization_id from Table Organization that the organizer is affiliated with. e.g 1**	**INT(10)**	**FOREIGN**	**10**	**NULL**	**NO**
4	**full_name**	**'full_name' represents the full name of the organizer representative. e.g Ahmad Ilman Wafi bin Mohd Azami**	**VARCHAR(100)**	**-**	**100**	**NULL**	**NO**
5	**position**	**'position' represents the organizer's position or role in the organization. e.g President**	**VARCHAR(50)**	**-**	**50**	**NULL**	**YES**
6	**phone_no**	**'phone_no' represents the mobile phone number of the organizer. e.g 012-1234567**	**VARCHAR(15)**	**-**	**15**	**NULL**	**YES**

5.TABLE CATEGORY

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL

1	category_id	'category_id' represents the unique category id that is generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	category_name	'category_name' represents the name of the opportunity category. e.g Workshop	VARCHAR(50)	UNIQUE	50	NULL	NO
3	description	'description' represents the details explaining the category scope.	VARCHAR(255)	-	255	NULL	YES
4	category_status	'category_status' represents the status of the category. 'A' represents Active and 'I' represents Inactive. e.g A	CHAR(1)	-	1	A	NO

6.TABLE EVENT

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	event_id	'event_id' represents the unique event id that is generated in	INT(10)	PRIMARY	10	0	NO

		auto-increment. e.g 1					
2	organizer_id	'organizer_id' represents the organizer_id from Table Organizer who created and manages the event. e.g 1	INT(10)	FOREIGN	10	NULL	NO
3	category_id	'category_id' represents the category_id from Table Category that classifies the event. e.g 1	INT(10)	FOREIGN	10	NULL	NO
4	title	'title' represents the title of the event or opportunity listing. e.g Software Engineering Workshop	VARCHAR(150)	-	150	NULL	NO
5	description	'description' represents the full event information and agenda presented by the organizer.	VARCHAR(500)	-	500	NULL	NO
6	event_mode	'event_mode' represents the delivery mode of the event ('Physical', 'Online', or 'Hybrid'). e.g Physical	VARCHAR(20)	-	20	Physical	NO

7	**location**	**'location' represents the physical venue or online platform link of the event. e.g Auditorium Block D**	**VARCHAR(150)**	**-**	**150**	**NULL**	**NO**
8	**start_datetime**	**'start_datetime' represents the start date and time of the event. e.g 2026-11-10 08:30:00**	**DATETIME**	**-**	**-**	**0000-00-00 00:00:00**	**NO**
9	**end_datetime**	**'end_datetime' represents the end date and time of the event. e.g 2026-11-10 17:00:00**	**DATETIME**	**-**	**-**	**0000-00-00 00:00:00**	**NO**
10	**reg_deadline**	**'reg_deadline' represents the registration closing date and time for the event. e.g 2026-11-05 23:59:59**	**DATETIME**	**-**	**-**	**0000-00-00 00:00:00**	**NO**
11	**capacity**	**'capacity' represents the maximum number of participants allowed for the event. e.g 100**	**INT(5)**	**-**	**5**	**0**	**NO**

12	**eligibility_criteria**	**'eligibility_criteria' represents the target participant eligibility requirements for the event. e.g Open to all computing students**	**VARCHAR(255)**	**-**	**255**	**NULL**	**YES**
13	**event_status**	**'event_status' represents the workflow status of the event ('D' for Draft, 'P' for Pending, 'A' for Approved, 'R' for Rejected, 'X' for Cancelled, 'C' for Completed). e.g P**	**CHAR(1)**	**-**	**1**	**P**	**NO**
14	**review_remarks**	**'review_remarks' represents the remarks or comments given by the administrator when approving or rejecting an event submission.**	**VARCHAR(255)**	**-**	**255**	**NULL**	**YES**

7.TABLE REGISTRATION

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL

1	**registration_id**	**'registration_id' represents the unique registration id generated in auto-increment each time a participant registers for an event. e.g 1**	**INT(10)**	**PRIMARY**	**10**	**0**	**NO**
2	**participant_id**	**'participant_id' represents the participant_id from Table Participant, which is the foreign key indicating the student who registered. e.g 1**	**INT(10)**	**FOREIGN**	**10**	**NULL**	**NO**
3	**event_id**	**'event_id' represents the event_id from Table Event, which is the foreign key indicating the event registered for. e.g 1**	**INT(10)**	**FOREIGN**	**10**	**NULL**	**NO**
4	**reg_datetime**	**'reg_datetime' represents the date and time when the participant registered for the event, generated automatically. e.g 2026-10-20 14:15:00**	**DATETIME**	**-**	**-**	**0000-00-00 00:00:00**	**NO**
5	**reg_status**	**'reg_status' represents the status of the registration ('R' for Registered,**	**CHAR(1)**	**-**	**1**	**R**	**NO**

		'C' for Cancelled, 'W' for Waitlisted, 'A' for Attended). e.g R					

8.TABLE BOOKMARK

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	bookmark_id	'bookmark_id' represents the unique bookmark id generated in auto-increment each time a participant bookmarks an event. e.g 1	INT(10)	PRIMARY	10	0	NO
2	participant_id	'participant_id' represents the participant_id from Table Participant who bookmarked the event. e.g 1	INT(10)	FOREIGN	10	NULL	NO
3	event_id	'event_id' represents the event_id from Table Event that was bookmarked by the participant. e.g 1	INT(10)	FOREIGN	10	NULL	NO

4	saved_datetime	'saved_datetime' represents the date and time when the event was bookmarked, generated automatically. e.g 2026-10-18 11:30:00	DATETIME	-	-	0000-00-00 00:00:00	NO

9.TABLE INTEREST

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	interest_id	'interest_id' represents the unique interest id generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	participant_id	'participant_id' represents the participant_id from Table Participant who maintains the interest. e.g 1	INT(10)	FOREIGN	10	NULL	NO
3	category_id	'category_id' represents the category_id from Table Category linked to the student's interest. e.g 1	INT(10)	FOREIGN	10	NULL	NO

4	interest_tag	'interest_tag' represents the specific interest topic keyword used for personalized event recommendations. e.g Artificial Intelligence	VARCHAR(50)	-	50	NULL	NO

10.TABLE SKILL

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	skill_id	'skill_id' represents the unique skill id generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	participant_id	'participant_id' represents the participant_id from Table Participant who possesses the skill. e.g 1	INT(10)	FOREIGN	10	NULL	NO
3	skill_name	'skill_name' represents the name of the skill possessed by the participant. e.g Python Programming	VARCHAR(50)	-	50	NULL	NO

4	proficiency_level	'proficiency_level' represents the skill proficiency level ('Beginner', 'Intermediate', or 'Advanced'). e.g Intermediate	VARCHAR(20)	-	20	Beginner	NO

11.TABLE NOTIFICATION

NO.	ATTRIBUTES	DESCRIPTION	TYPE	KEY	LENGTH	DEFAULT	NULL
1	notification_id	'notification_id' represents the unique notification id generated in auto-increment. e.g 1	INT(10)	PRIMARY	10	0	NO
2	account_id	'account_id' represents the account_id from Table Account indicating the user who receives the notification. e.g 1	INT(10)	FOREIGN	10	NULL	NO
3	event_id	'event_id' represents the event_id from Table Event that triggered the notification. e.g 1	INT(10)	FOREIGN	10	NULL	YES
4	title	'title' represents the headline title of the notification. e.g Upcoming Registration Deadline	VARCHAR(100)	-	100	NULL	NO

5	**message**	**'message' represents the full content of the notification message.**	**VARCHAR(255)**	**-**	**255**	**NULL**	**NO**
6	**notification_type**	**'notification_type' represents the category of notification ('Reminder', 'Deadline', 'Cancellation', or 'Recommendation'). e.g Deadline**	**VARCHAR(30)**	**-**	**30**	**NULL**	**NO**
7	**is_read**	**'is_read' represents the read status of the notification ('Y' for Read, 'N' for Unread). e.g N**	**CHAR(1)**	**-**	**1**	**N**	**NO**
8	**sent_datetime**	**'sent_datetime' represents the date and time when the notification was sent, generated automatically. e.g 2026-11-01 08:00:00**	**DATETIME**	**-**	**-**	**0000-00-00 00:00:00**	**NO**